

A FINAL REPORT ON
PREDICTING STOCK MARKET TRENDS USING
MACHINE LEARNING AND DEEP LEARNING

SUBMITTED TO
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BACHELOR OF ENGINEERING
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SUBMITTED BY

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UNDER THE GUIDANCE OF

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CERTIFICATE

This is to certify that the project entitled

“PREDICTING STOCK MARKET TRENDS USING MACHINE LEARNING AND DEEP LEARNING”

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ABSTRACT

Stock Request is trading platform where different investors trade and purchase shares according to stock vacuity. Stock request ups and campo goods the profit of stakeholders. If request prices going up with available stock also stakeholders get profit with their bought stocks. In other case, if request going down with available stock prices also stakeholders have to face losses. Buyers buy stocks with low prices and vend stocks at high prices and try to get huge profit. Also, merchandisers vend their products at high prices for profit purpose. Stock request work as safe platform among merchandisers and buyers .Stocks are conceivably the most popular fiscal instrument constructed for erecting wealth and are the centre piece of any investment portfolio.

The advances in trading technology has opened up the requests so that currently nearly anybody can enjoy stocks. From last many decades, there seen explosive increase in the average person's interest for stock request. In a financially explosive request, as the stock request, it's important to have a veritably accurate vaticination of a unborn trend. Because of the fiscal extremity andre-cording gains, it's mandatory to have a secure vaticination of the values of the stocks. Predicting anon-linear signal requires progressive algorithms of machine literacy with help of Artificial Intelligence (AI).

In Stock Market Prediction, the end is to prognosticate the unborn value of the fiscal stocks of a company. The recent trend in stock request vaticination technologies is the use of machine literacy which makes prognostications grounded on the values of current stock request indicators by training on their former values. Machine learning itself employs different models to make vaticination easier and authentic. The paper focuses on the use of Retrogression and Arima grounded Machine literacy to prognosticate stock values. Factors considered are open, close, low, high and volume.

Keywords given to the voice recognition features are as follows:-

- **Forecasting**
- **Data analysis**
- **Prediction,**
- **Text mining**

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LIST OF ABBREVIATIONS

ABBREVIATION	ILLUSTRATION
VPN	Virtual Private Network
IP	Internet Protocol
IDS	Intrusion Detection System
TCP	Transmission Control Protocol

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CHAPTER 01

01 Introduction

1.1 Overview

The fast progress in digital information acquisition has crystal rectifier to the invasive quantity of knowledge hold on in databases, information warehouses, or different kinds of knowledge repositories. though valuable info could also be concealment behind the info, the overwhelming information volume makes it tough for citizenry to extract them while not powerful tools. straightforward and fast availableness to news info wasn't attainable till the start of the last decade. during this age of data, news is currently simply accessible, as content suppliers and content locators like on-line news services have up on the planet Wide net. Continuous availableness of additional news articles in digital type, the newest developments in linguistic communication process (NLP) and therefore the availableness of quicker computers cause the question the way to extract additional info out of reports articles.

Financial analysts United Nations agency invest available markets typically don't seem to be attentive to the exchange behavior. they're facing the matter of stock commercialism as they are doing not recognize that stocks to shop for and that to sell so as to achieve additional profits. of these users recognize that the progress of the exchange depends a great deal on relevant news and that they ought to deal daily with immense quantity of data. they need to investigate all the news that seems on newspapers, magazines and alternative matter resources. however analysis of such quantity of economic news and articles so as to extract helpful information exceeds human capabilities. Text mining techniques will facilitate them mechanically extracting the helpful information out of matter resources.

1.2 Motivation

In the existing system there's AN solely Current worth of stock value is done. That mean we are able to take solely the front and back read of the item and store it.

1.3 Problem Definition and Objectives

Difficult to determine STOCK value prediction. has been at focus for years since it will yield vital growth by doing varied investment. Predicting the exchange isn't a straightforward task, chiefly as a consequence of the about to stochastic process behavior of a stock statistic.

Objectives:

In Stock Market Prediction, the aim is to predict the long run worth of the money stocks of an organization. The recent trend available market prediction technologies is that the use of machine learning that makes predictions supported the values of current exchange

1.4 Project Scope & Limitations

The exchange prediction has attracted abundant attention from world additionally as business. thanks to the non-linear, volatile and complicated nature of the market, it's quite tough to predict. because the stock markets grow larger, additional investors concentrate to develop a scientific approach to predict the exchange. Since the exchange is incredibly sensitive to the external info, the performance of previous prediction systems is restricted by just considering the standard stock information.

1.5 Methodologies of drawback finding

In this project, we've got in contestible a machine learning approach (deep learning) to predict exchange trend mistreatment. Results show however history information has been accustomed predict stock movement with affordable accuracy. Also, with T take a look at result analysis we are able to conclude that Arima performs higher compared to statistical regression. For this implementation, we are able to conclude that if we have a tendency to incorporate all the factors that have an effect on performance of the stock and feed them to with correct information preprocessing and filtering, once coaching the network we are going to be ready to have a model could which might predict stock momentum additional accurately and exactly for the higher plan of stock worth so companies may have accumulated profit magnitude relation as compared to what's may be going presently at that point. this may conjointly cause

additional transparency concerning stock because it are easier for companies to investigate loses and deliver the goods nice success.

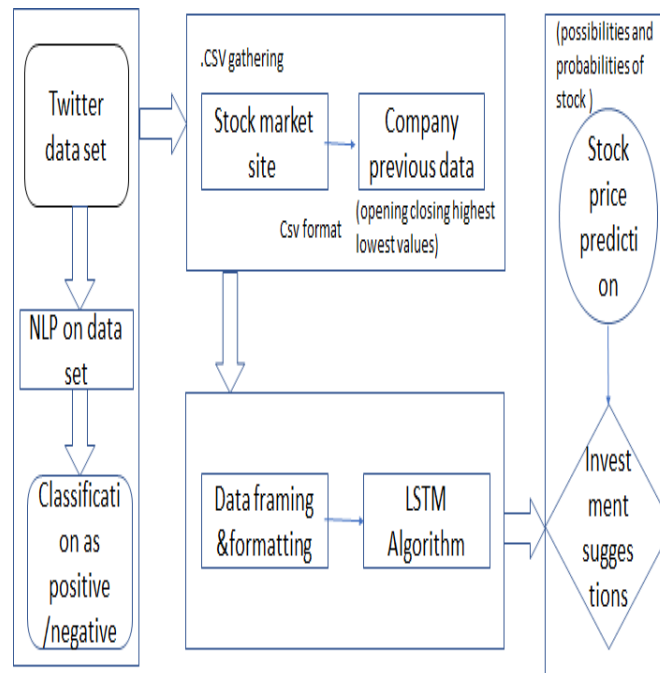


Fig. 1: Working process of the system

CHAPTER 02

02 Literature Survey

This section describes related work. We have identified related work in two different areas: technology and finance. Kim and Han constructed a discretized model of the function for predicting stock indexes). The data used in their study include technical indicators and the direction of daily changes in the Korea Stock Index (KOSPI). They used data containing a sample of 2928 trading days from January 1989 to December 1998 to report selected characteristics and formulas. They also applied feature discretization optimization as a technique similar to dimensionality reduction. The strength of their work is the implementation of GA to optimize ANN. First, the number of hidden layer input features and processing elements is 12, which cannot be adjusted. Another limitation lies in the learning process of ANN, where the author focuses only on two elements of optimization. Although they still believed that GA had great potential for optimizing the discretization of functions. The initialized feature pool is related to the selected nature. Qiu and Song also presented a solution in for predicting the direction of the Japanese stock market based on an optimized model. In this work, the author uses a genetic algorithm with a model based on an, which he calls a hybrid GAANN model. Piramuthu conducted a thorough evaluation of various methods of feature selection for data mining applications. He used records, which are credit approval data, credit default data, web traffic data, Tam and Kiang data, and compared how various feature selection methods optimized the performance of decision trees. The methods he compared to select features included stochastic distance measures: Batachariya measure, Matsushita measure, divergence measure, Mahalanobis distance measure, and Patrick Fisher measure. For interclass distance measures: Minkowski distance measures, city block distance measures, Euclidean distance measures, Chebicev distance measures, and non-linear (Parzen and hypersphere kernel) distance measures. Hassan and Nath applied the Hidden Markov Model (HMM) to stock market forecasts for stock prices off our different airlines. Reduces the state of the model to four states: open price, close price, highest price, and lowest price. The strength of this white paper is that this approach does not require expertise to build a predictive model. This task is limited within the aviation industry and is interpreted on very small

datasets, but it may not be a predictive model with general validity. One of the working approaches to stock market forecasting can be used for comparison work. Predicting of Stock Market trends using Machine Learning and deep learning 4 | PVG college of Engineering The author has selected a maximum of 2 years as the date range for training and test datasets. This provided a reference to the date range of the evaluation part. For optimization, the author also used a rough set (RS) for attribute reduction. Rough sets were used to reduce the dimension of stock price trends. It was also used to determine the structure of the wavelet network. The dataset for this work consists of five well-known stock market indices: (1) SSE Composite Index (China), (2) CSI 300 Index (China), (3) All Common Indexes (Australia), (4). It is composed of Nikkei. 225 Index (Japan) and (5) Dow Jones Index (USA). The model's valuation is based on various stock market indices, and the results were convincing because of its general validity. Using rough sets to optimize feature dimensions before processing reduces computational complexity. However, in the discussion part, the author emphasized only parameter adjustments and did not identify weaknesses in the model itself. In the meantime, I also noticed that the index was beige valuated. If you apply the same model to a particular stock, you may not get the same performance. Siri nano and Cont. were used in a deep learning solution trained on the universal functions of financial markets. The dataset used consisted of trading records for all transactions and cancellation of orders for approximately 1000 NASDAQ shares via stock exchange purchase orders. The NN consists of three layers with Arima units and finally a feed forward layer with a rectified linear unit (ReLU), using a stochastic gradient descent(SGD) algorithm for optimization. Their universal model was able to generalize and cover strains other than those in the training data. They mentioned the benefits of the universal model, but the cost of training was still high. In the meantime, it's unclear if the mysterious programming of deep learning algorithms contaminates useless functions when data is fed into the model. The authors found that it was better to perform .

CHAPTER 03

3.1 Assumptions and Dependencies

It is assumed that answer information are created accessible for the project in some part of its completion. Until then, check information are used for providing the demo for the shows. it's assumed that the user is accustomed to a web browser and additionally accustomed to handling the keyboard and mouse. Since the applying could be a net based mostly application there's a necessity for the web browser. can it'll be assumed that the users will possess tight net property.

3.2 Functional needs

3.2.1 System Feature 1(Functional Requirement)

This section describes the useful needs of the applying and therefore the options it provides. System options area unit represented thoroughly to assist the long run extension and testing of the system. options declared here area unit already components of the enforced system therefore no prioritization is required. Priority is required for options to be developed which will be other to the present document later.

3.2.2 System Feature2 (Functional Requirement)

- Profiles:

To manage your profiles, think again the web site so you'll either select one among the stock file for prediction.

- Options:

A wide style of choices is obtainable to the user, together with interface choices, select Stock file, select algorithms.

3.3 External Interface needs (If Any)

3.3.1 User Interfaces

User interface of this program is that the common windows interface, nothing further is needed. the event of securities market worth prediction to attach stock interface ought to be intuitive, such ninety-nine.9% of all new users area unit ready to use this application with none help.

3.3.2 Hardware Interfaces

The hardware ought to have following specifications:

- Ability to scan gallery
- Ability to exchange information over network
- bit screen for convenience
- keyboard (in case touchpad not available)
- Continuous power provide
- Ability to attach to network
- Ability to require input from user
- Ability to validate user

3.3.3 Software Interfaces

The computer code interfaces area unit specific to the target different user's Development of Stock worth Prediction supported Machine Learning Approach computer code systems.

3.3.4 Communication Interfaces

- All the program's options area unit accessible offline. you simply want a web association to transfer the applying, update it and (optionally) to urge some target pictures for your personal profile.
- Our Project belongs to mechanical man based mostly, therefore connecting user at on-line with request and response form. For that HTTP protocol we tend to area unit aiming to U.S.. that's provided by PHP net services.

- HTTP protocol: The machine-readable text Transfer protocol is Associate in Nursing application protocol for distributed, collaborative, and hypermedia system info systems. HTTP is that the foundation of information communication for the World Wide Web. machine-readable text is structured text that uses logical links (hyperlinks) between nodes containing text.

3.4 Non-functional needs

3.4.1 Performance needs

System will turn out results quicker on 4GB of RAM.

- It may take longer for peak masses at main node.
- The system are accessible 100% of the time. Once there's a fatal error, the system can provide understandable feed back to the user.

3.4.2 Safety needs

Only directors have access to the information of every individual user.

- All information are backed-up each day mechanically and additionally the supervisor will backup the information as a operate for him.
- This makes it easier to put in and updates new practicality if needed.
- For the security purpose backup of the information should be needed.

3.4.3 Security needs

Our System is being developed in python. Python is Associate in Nursing taken high-level general programing language. Its style philosophy emphasizes code readability with its use of great indentation. Its language constructs further as its object-oriented approach aim to assist programmers write clear, logical code for little and large-scale comes.

- For the protection functions and to avoid unlawful use of the system, whereas exploitation this application user should do following things:
- At the time of deploying this computer code user ought to register to system.
- To use computer code user, ought to login and logout whenever.

3.4.4 Software Quality Attributes

The system considers following non-functional needs to supply higher functionalities and usage of system.

- 1. Availability:** The system shall be accessible throughout twenty-four hours of each day.
- 2. Usability:** The system is intended keeping in mind the usability problems considering the end-users World Health Organization area unit developers/programmers. It provides elaborate facilitate which might cause higher and quicker learning. Navigation of system is straightforward.
- 3. Consistency:** Uniformity in layout, screens, Menus, colors theme, format.
- 4. Performance:** The performance of the system ought to be quick and as per user demand. From this method we'll get expected outcome in less time and fewer house since potency is higher. Speed is completely looking on the response of the information and association sort.
- 5. Extendibility:** bar within the system ought to be exhausted the system by that we tend to create changes within the system in a while.
- 6. Reusability:** Files of any sort are often employed by the system for any variety of times throughout transformation.
- 7. Reliability:** Protection of information from malicious attack or unauthorized access.
- 8. Security:** The system provides security to the haphazardly generated non-public key by playing coding thereto for encrypting patient information and therefore protects from different nodes within the network. The network is free from malicious node and misbehaving node attacks.
- 9. Reliability:** Our system will offer user Associate in Nursing economical search whenever. that the user will reliable on the system. as a result of system will guarantee user to supply his/her interested information on every occasion in slightest of your time

3.5 System Requirements

3.5.1 Database Requirements

Google Stock Price dataset

3.5.2 Software Requirements (Platform Choice)

Python:

Python defines an in-built module platform that provides system information. The Platform module is used to retrieve as much possible information about the platform on which the program is being currently executed.

Python is an interpreted high-level general-purpose programming language. Its design philosophy emphasizes code readability with its use of significant indentation. Its language constructs as well as its object-oriented approach aim to help programmers write clear, logical code for small and large-scale projects.

Python is dynamically-typed and garbage-collected. It supports multiple programming paradigms, including structured (particularly, procedural), object-oriented and functional programming. It is often described as a "batteries included" language due to its comprehensive standard library.

Guido van Rossum began working on Python in the late 1980s, as a successor to the ABC programming language, and first released it in 1991 as Python 0.9.0.[32] Python 2.0 was released in 2000 and introduced new features, such as list comprehensions and a garbage collection system using reference counting. Python 3.0 was released in 2008 and was a major revision of the language that is not completely backward-compatible. Python 2 was discontinued with version 2.7.18 in 2020.

Python consistently ranks as one of the most popular programming languages

3.5.3 Hardware Requirements

System : Pentium IV 2.4 GHz.

Hard Disk : 40 GB.

Floppy Drive : 1.44 Mb.

Monitor : 15 VGA Colour.

Mouse : Logitech.

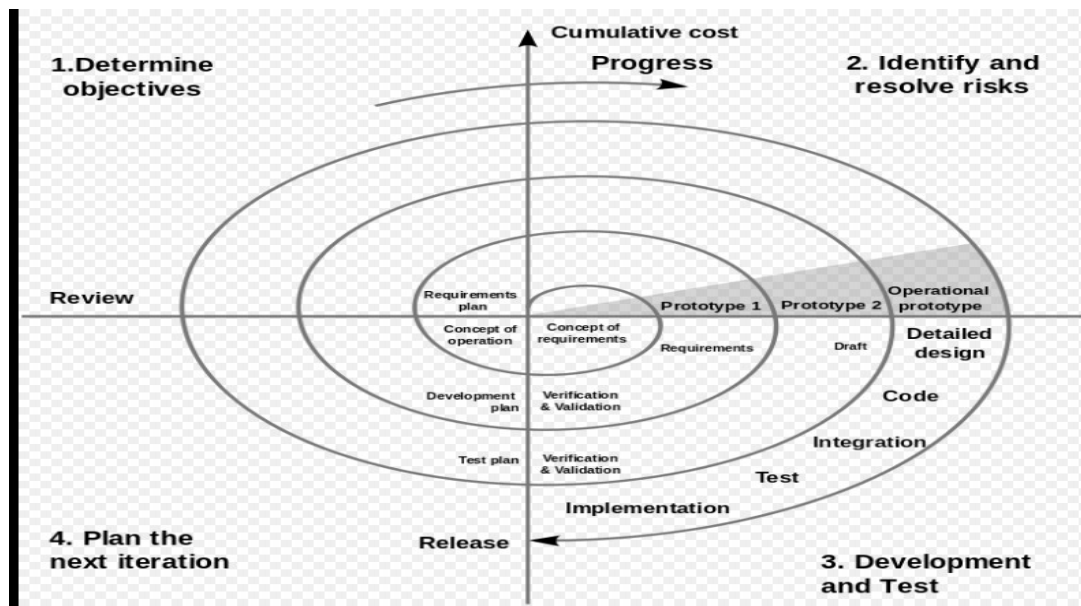
Ram : 512 Mb.

3.6 Analysis Models: SDLC Model to be applied

Spiral model is one of the most important Software Development Life Cycle models, which provides support for **Risk Handling**. In its diagrammatic representation, it looks like a spiral with many loops. The exact number of loops of the spiral is unknown and can vary from project to project. **Each loop of the spiral is called a Phase of the software development process.** The exact number of phases needed to develop the product can be varied by the project manager depending upon the project risks. As the project manager dynamically determines the number of phases, so the project manager has an important role to develop a product using spiral model.

The Radius of the spiral at any point represents the expenses(cost) of the project so far, and the angular dimension represents the progress made so far in the current phase.

Below diagram shows the different phases of the Spiral Model:



Each phase of Spiral Model is divided into four quadrants as shown in the above figure. The functions of these four quadrants are discussed below-

1. Objectives determination and identify alternative solutions:

Requirements are gathered from the customers and the objectives are identified, elaborated and analyzed at the start of every phase. Then alternative solutions possible for the phase are proposed in this quadrant.

2. Identify and resolve Risks:

During the second quadrant all the possible solutions are evaluated to select the best possible solution. Then the risks associated with that solution is identified and the risks are resolved using the best possible strategy. At the end of this quadrant, Prototype is built for the best possible solution.

3. Develop next version of the Product:

During the third quadrant, the identified features are developed and verified through testing. At the end of the third quadrant, the next version of the software is available.

4. Review and plan for the next Phase: In the fourth quadrant, the Customers evaluate the so far developed version of the software. In the end, planning for the next phase is started.

CHAPTER 04

04 System Design

4.1 System Architecture

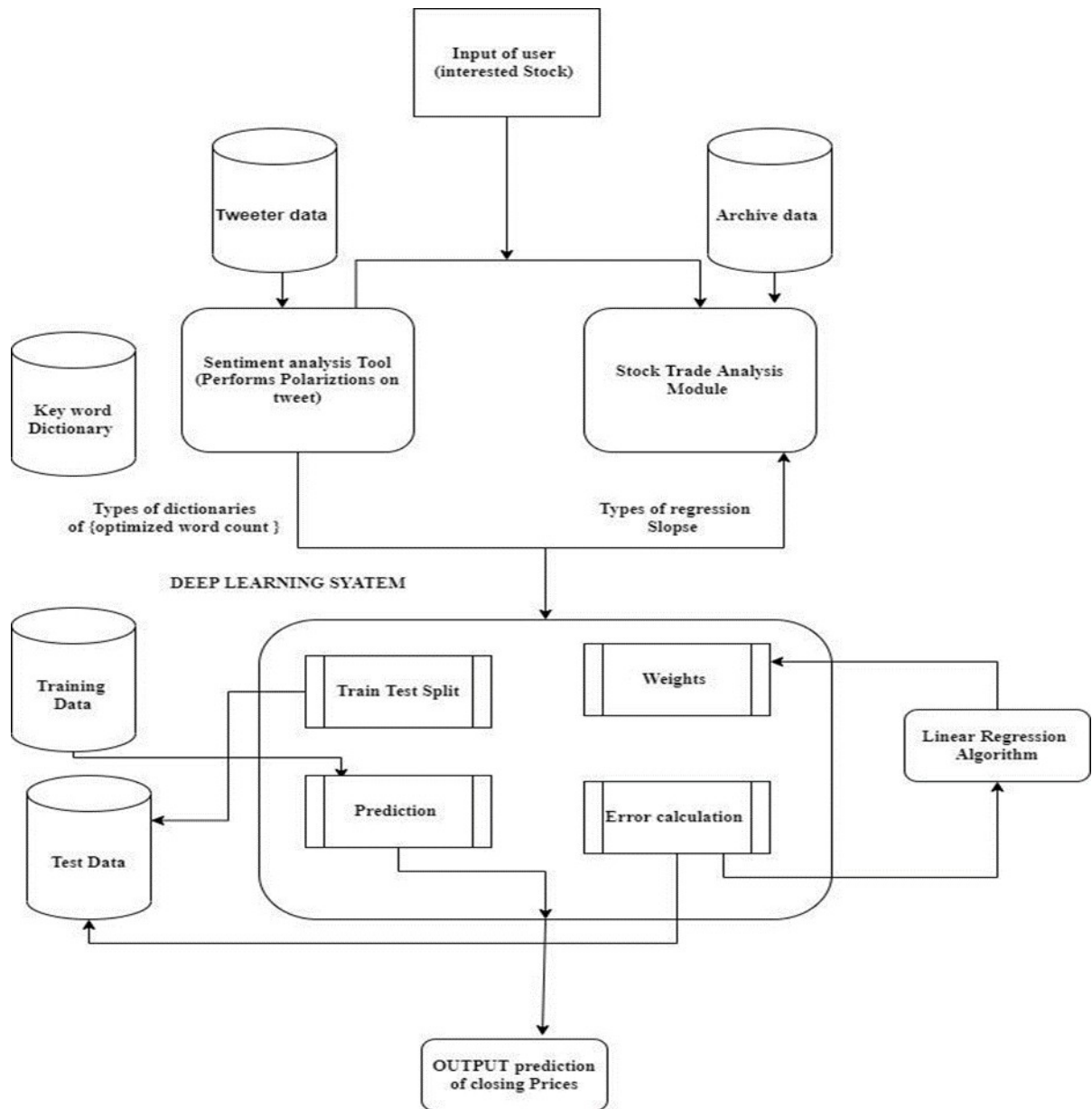


Fig 01.1 System Architecture

4.2 Data Flow Diagram

Data flow diagram level

0



Fig 4.2.1: DFD Level 0

Data flow diagram level 1

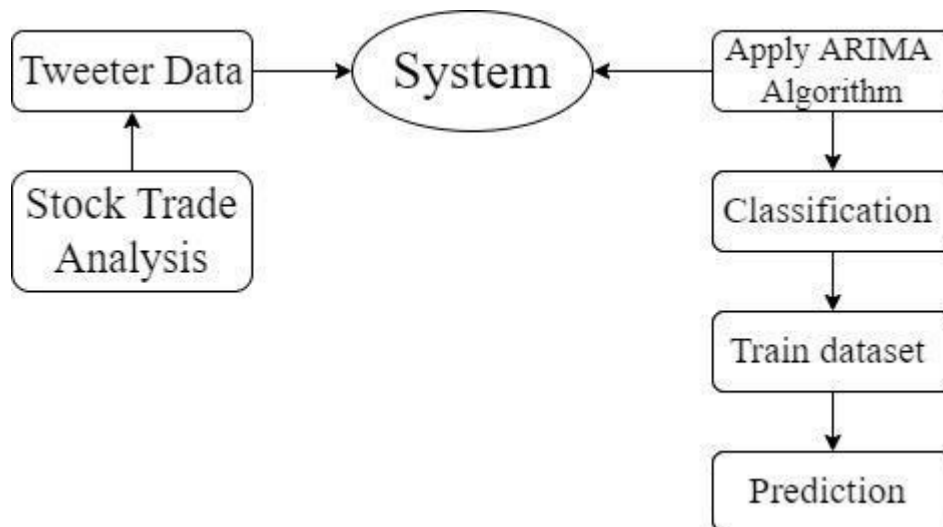
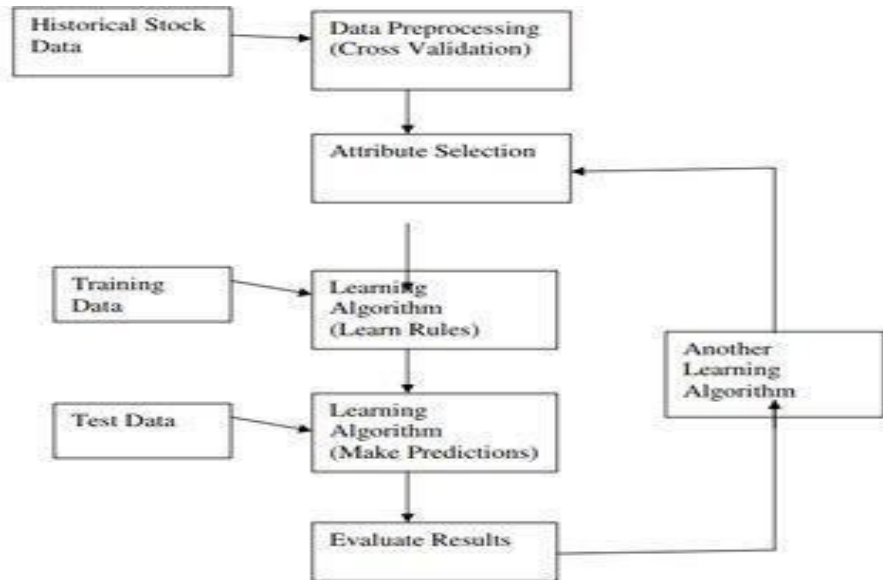


Fig 4.2.2: DFD Level 1

Data flow diagram level 2**Fig 4.2.3: DFD Level 2**

4.3 Entity Relationship Diagram

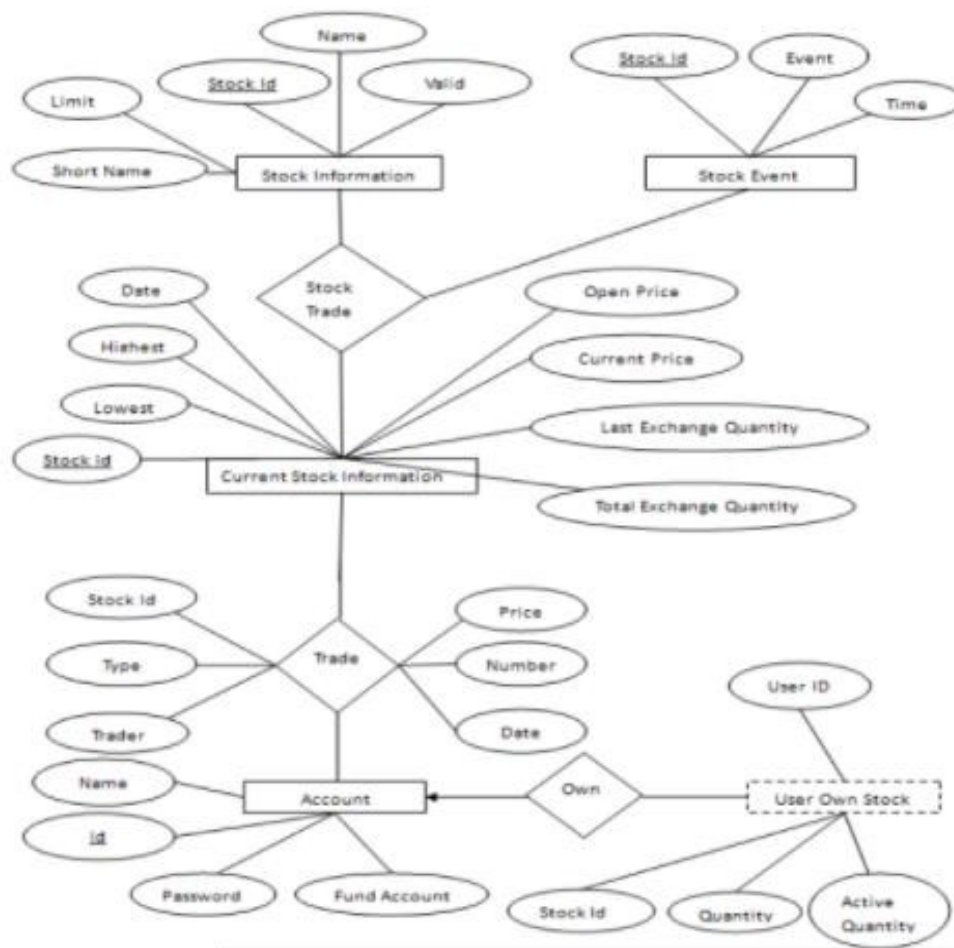


Fig 4.3 Entity Relationship Diagram

4.4 UML Diagrams

4.4.1 Use case diagram

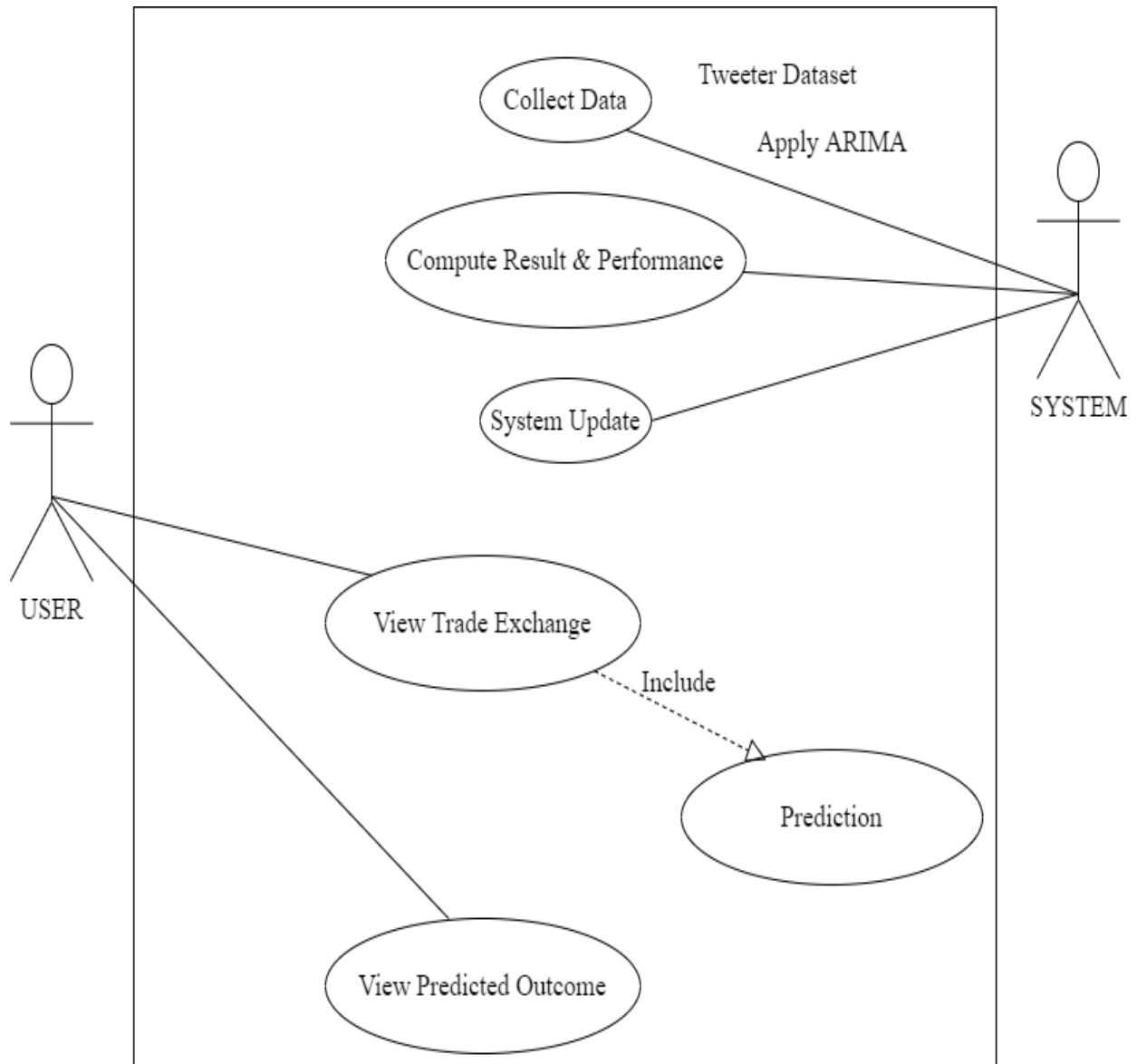


Fig 4.4.1: Use case diagram

4.4.2 Sequence Diagram

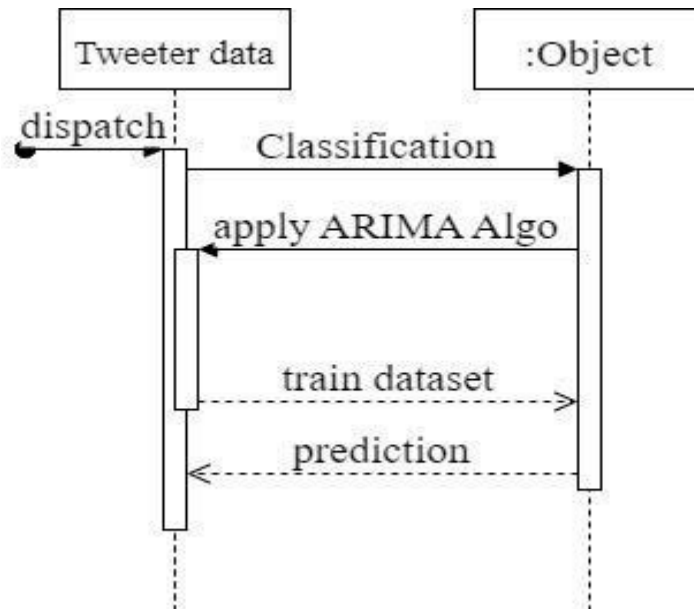


Fig 4.4.2: Sequence diagram

4.4.3 Activity Diagram

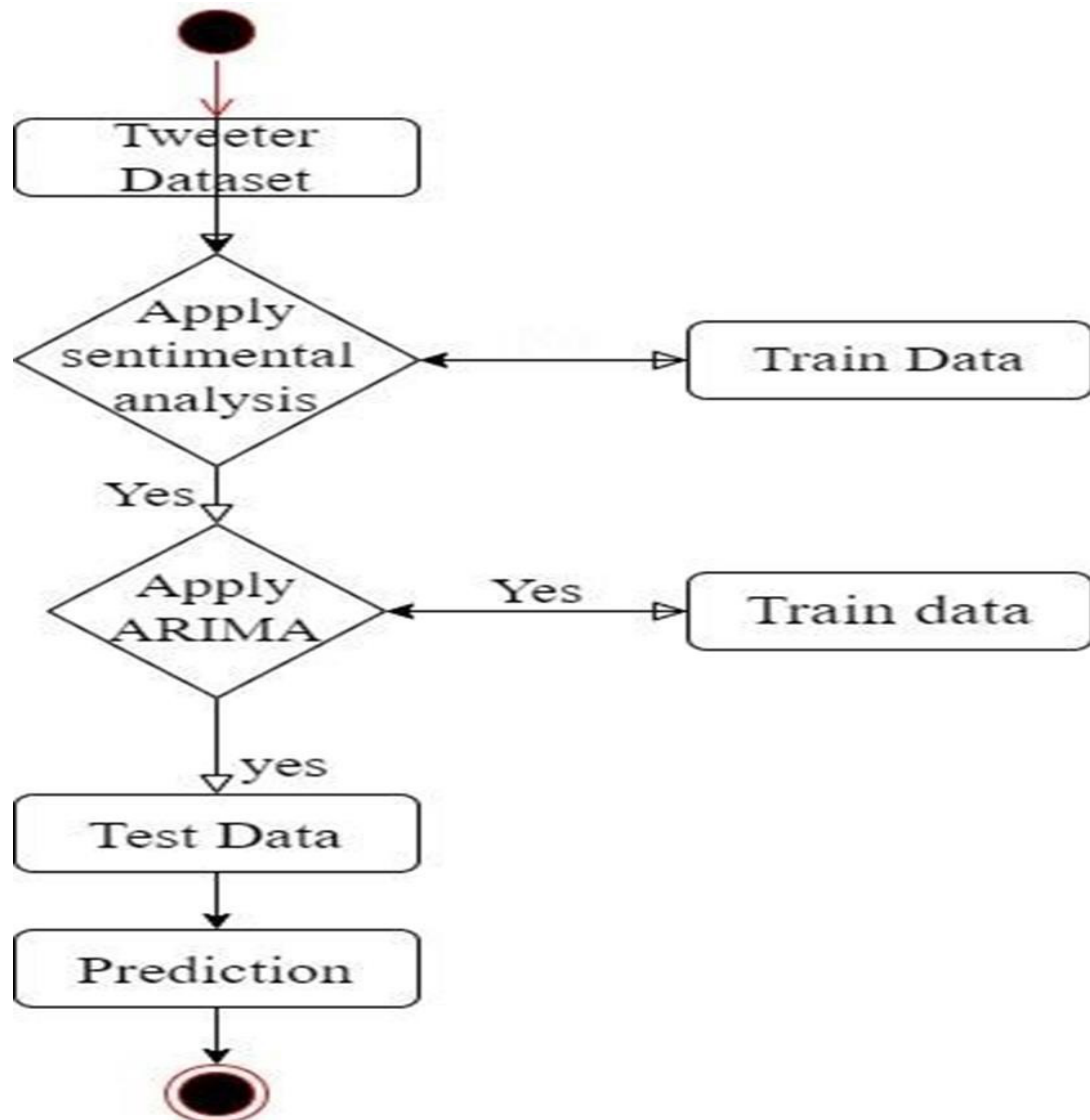


Fig 4.4.3: Activity diagram

4.4.4 Class Diagram

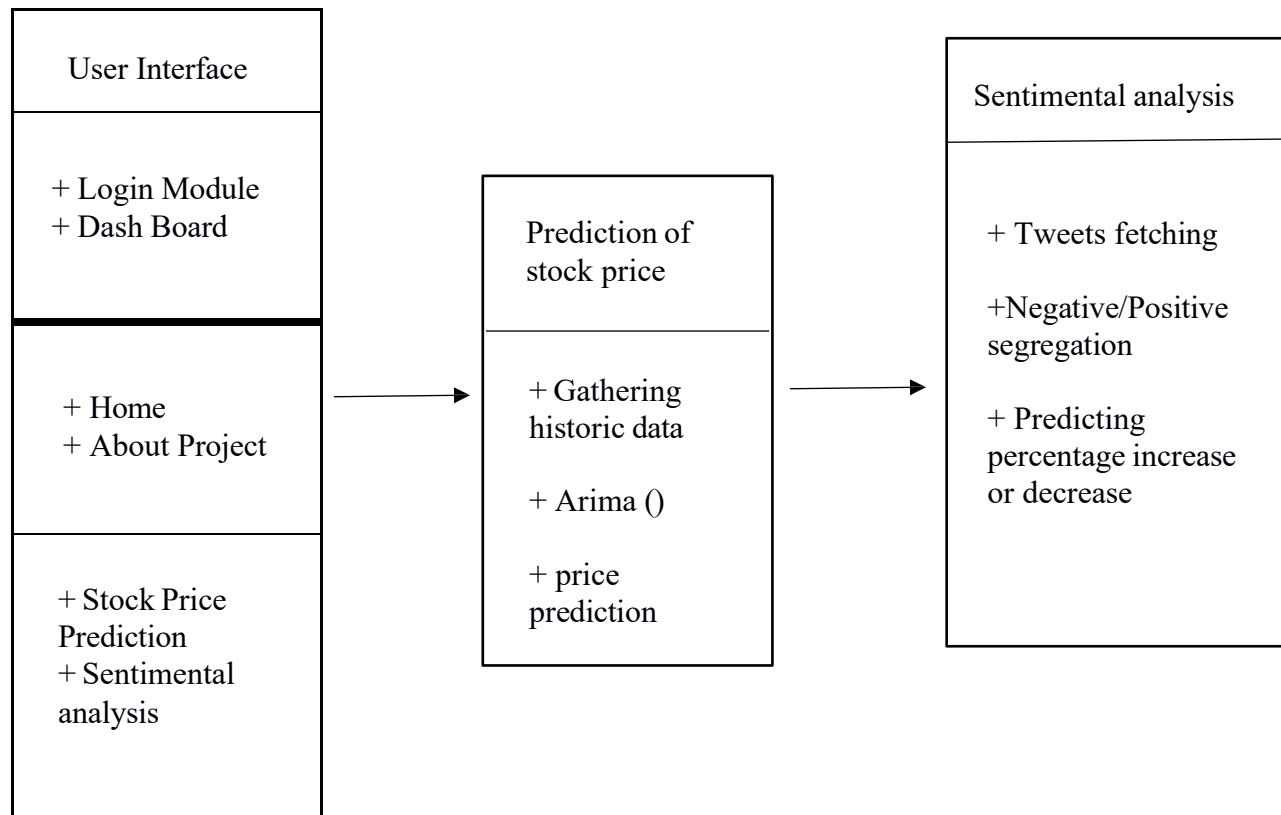


Fig.4.4.4 Class diagram

CHAPTER 05

05 Project Plan

5.1 Project Estimate

❖ Cost Estimate

The project cost can be found using any one of the model.

COCOMO-1 Model

COCOMO-2 Model

Model -1: The basic COCOMO model computes software development efforts as a function of program size expressed in estimated lines of code.

Model-2: The intermediate COCOMO model computes software development efforts as a function of program size and a set of cost drivers that include subjective assessment of the product, hardware, personnel, project attributes

Model-3: The advanced COCOMO model incorporates all characteristics of the intermediate version with an assessment of the cost drivers impact on each step of the software engineering process. Following is the basic COCOMO -2 model.

Software Project	A(b)	B(b)	C(b)	D(b)
Organic	2.4	1.05	2.5	0.38
Semi-detached	3.0	1.22	2.5	0.35
Embedded	3.6	1.20	2.5	0.32

The basic COCOMO -2 model equations take form:

$$E = A(b)KLOC^{B(b)}$$

$$D = C(b)E^{D(b)}$$

Where E is the effort applied in person months. D is development time in chronological month. KLOC is estimated number of delivered lines of code for the project. This project can be classified as Semidetached software project. The rough estimate of number of lines of this project is 9.072k. Applying the above formula

$$E = 3.0 * (9.072)^{1.22}$$

$$= 44.20 \text{ person-months}$$

$$D = 2.5 * 44^{.35}$$

$$= 9.40 \text{ months}$$

Hence according COCOMO -2 model the time required for completion of the project is 9 (~9.40) months.

❖ Cost of Project:

Equation for calculation of cost of project using COCOMO - 2 model is:

$$C = D * C_p$$

Where,

C = Cost of project

D = Duration in month

C_p = Cost incurred per person-month, C_p = Rs.5000/- (per person-month) (approx.)

$$C = 9 * 5000$$

$$= 45000/-$$

Hence according COCOMO - 2 model the cost of project is 45000/-(approx.)

❖ Time Estimates

The time estimate of this project is approximate 11 months.

2. Project Resources

Well configured Laptop, eclipse IDE, 2 GHZ CPU speed, 2 GB RAM, Internet connection

5.2 Risk Management

Risk Management w.r.t. NP Hard analysis:

1. In appropriate dataset -To overcome this risk we are trying to use well organized and complete dataset.
2. Security- To overcome and improving security we use multilevel security like access permissions of user.

Risk Identification:

1. Have top software and customer managers formally committed to support the project?

Ans-Not applicable.

2. Are end-users enthusiastically committed to the project and the system/product to be built?

Ans-Not known at this time.

3. Are requirements fully understood by the software engineering team and its customers?

Ans-Yes

4. Have customers been involved fully in the definition of requirements?

Ans-Not applicable

5. Do end-users have realistic expectations?

Ans-Not applicable

6. Does the software engineering team have the right mix of skills?

Ans-yes

7. Are project requirements stable?

Ans-Not applicable

8. Is the number of people on the project team adequate to do the job?

Ans-Not applicable

9. Do all customer/user constituencies agree on the importance of the project and on the requirements for the system/product to be built?

Ans-Not applicable

Risk Analysis:

The risks for the Project can be analyzed within the constraints of time and quality

Risk Table:

ID	Risk Description	Probability	Schedule	Quality	Overall
1	correctness	low	Low	high	Low
2	Availability	high	Low	high	high

Risk Probability definitions:

Probability	Value	Description
High	Probability of occurrence is	$> 75\%$
Medium	Probability of occurrence is	$26 - 75\%$
Low	Probability of occurrence is	$< 25\%$

Risk Impact definitions:

Impact	Value	Description
Very high	> 10%	Schedule impact or Unacceptable quality
High	5 – 10%	Schedule impact or Some parts of the project have low quality
Medium	< 5%	Schedule impact or Barely noticeable degradation in quality Low Impact on schedule or Quality can be incorporated

Risk Mitigation Risk Monitoring and Risk Management**1. Risk Mitigation:**

If a software team adopts a proactive approach to risk, avoidance is always the best strategy. This is achieved by developing a plan for *risk mitigation*. To mitigate this risk, you would develop a strategy for reducing turnover. Among the possible steps to be taken are:

- Meet with current staff to determine causes for turnover (e.g., poor working conditions, low pay, competitive job market).
- Mitigate those causes that are under your control before the project starts.
- Once the project commences, assume turnover will occur and develop techniques to ensure continuity when people leave.
- Organize project teams so that information about each development activity is widely dispersed.
- Define work product standards and establish mechanisms to be sure that all models and documents are developed in a timely manner.
- Conduct peer reviews of all work (so that more than one person is “up to speed”).
- Assign a backup staff member for every critical technologist.

2. Risk Monitoring

As the project proceeds, *risk-monitoring* activities commence. The project manager monitors factors that may provide an indication of whether the risk is becoming more or less likely. In the case of high staff

turnover, the general attitude of team members based on project pressures, the degree to which the team has jelled, interpersonal relationships among team members, potential problems with compensation and benefits, and the availability of jobs within the company and outside it are all monitored.

3. Risk Management

Risk management and contingency planning assumes that mitigation efforts have failed and that the risk has become a reality. Continuing the example, the project is well under way and a number of people announce that they will be leaving. If the mitigation strategy has been followed, backup is available, information is documented, and knowledge has been dispersed across the team. In addition, you can temporarily refocus resources (and readjust the project schedule) to those functions that are fully staffed, enabling newcomers who must be added to the team to “get up to speed.”

CHAPTER 06

06 Project Implementation

6.1 Overview of Project Modules

In this project, we have demonstrated a machine learning approach (deep learning) to predict stock market trend using different networks. Results show how history data has been used to predict stock movement with reasonable accuracy. Also, with T test result analysis we can conclude that Arima performs better in comparison to Linear Regression. For this implementation, we can conclude that if we incorporate all the factors that affect performance of the stock and feed them with proper data preprocessing and filtering, after training the network we will be able to have a model which can predict stock momentum more accurately and precisely for the better idea of stock value so that firms may have increased profit ratio as compared to what is might be going currently at that time. This will also lead to more transparency regarding stock as it will be easier for firms to analyze loses and achieve great success.

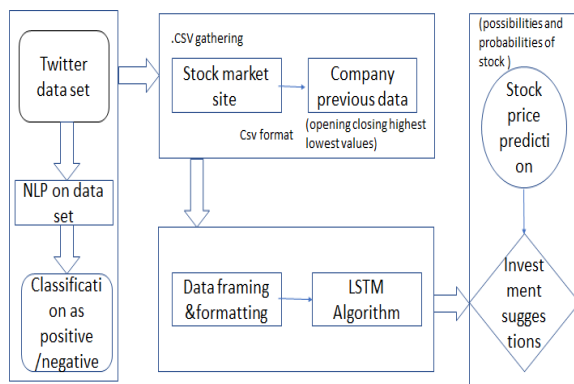


Fig. 1: Working process of the system

6.2 Tools and Technologies Used

Technology Used:

PYTHON

Python is a high-level, interpreted, interactive and object-oriented scripting language. Python is designed to be highly readable. It uses English keywords frequently where as other languages use punctuation, and it has fewer syntactical constructions than other languages.

- **Python is Interpreted** – Python is processed at runtime by the interpreter. You do not need to compile your program before executing it. This is similar to PERL and PHP.
- **Python is Interactive** – You can actually sit at a Python prompt and interact with the interpreter directly to write your programs.
- **Python is Object-Oriented** – Python supports Object-Oriented style or technique of programming that encapsulates code within objects.
- **Python is a Beginner's Language** – Python is a great language for the beginner-level programmers and supports the development of a wide range of applications from simple text processing to WWW browsers to games.

History of Python

Python was developed by Guido van Rossum in the late eighties and early nineties at the National Research Institute for Mathematics and Computer Science in the Netherlands.

Python is derived from many other languages, including ABC, Modula-3, C, C++, Algol-68, Small Talk, and Unix shell and other scripting languages.

Python is copyrighted. Like Perl, Python source code is now available under the GNU General Public License (GPL).

Python is now maintained by a core development team at the institute, although Guido van Rossum still holds a vital role in directing its progress

Python Features

Python's features include –

- **Easy-to-learn** – Python has few keywords, simple structure, and a clearly defined syntax. This allows the student to pick up the language quickly.
- **Easy-to-read** – Python code is more clearly defined and visible to the eyes.
- **Easy-to-maintain** – Python's source code is fairly easy-to-maintain.
- **A broad standard library** – Python's bulk of the library is very portable and cross-platform compatible on UNIX, Windows, and Macintosh.
- **Interactive Mode** – Python has support for an interactive mode which allows interactive testing and debugging of snippets of code.
- **Portable** – Python can run on a wide variety of hardware platforms and has the same interface on all platforms.
- **Extendable** – You can add low-level modules to the Python interpreter. These modules enable programmers to add to or customize their tools to be more efficient.
- **Databases** – Python provides interfaces to all major commercial databases.
- **GUI Programming** – Python supports GUI applications that can be created and ported to many system calls, libraries and windows systems, such as Windows MFC, Macintosh, and the X Window system of Unix.
- **Scalable** – Python provides a better structure and support for large programs than shell scripting.

6.3 Algorithm Details

1. Auto Regressive Integrated Moving Average (ARIMA)

- Time series forecasting models are the models that are capable to predict future values based on previously observed values
- Time series forecasting is widely used for non stationary data
- Non stationary data are called the data whose statistical properties e.g. the mean and standard deviation are not constant over time but instead, these metrics vary over time

Breaking down these terms we get :

- AR: < Auto Regressive > means that the model uses the dependent relationship between an observation and some predefined number of lagged observations (also known as “time lag” or “lag”).
- I: < Integrated > means that the model employs differencing of raw observations (e.g. it subtracts an observation from an observation at the previous time step) in order to make the time series stationary.
- MA: < Moving Average > means that the model exploits the relationship between the residual error and the relation between the residual error and the observation

Steps for implementation are as follows:

1. Visualize the time series Data
2. Make the time series data stationary
3. Plot the correlation and Auto-Correlation Chart
4. Construct the ARIMA model based on the data
5. Use the model to make prediction

Mathematical Equation For ARIMA is as follows :

$$Y_t = \mu + \epsilon_t$$

$$Y_t = \mu + \epsilon_t - 1 + \mu + \epsilon_t - 1 + \mu + \epsilon_t$$

$$\mu + \epsilon_t - 1 + \mu + \epsilon_t - 1 + \mu + \epsilon_t$$

$$\mu + \epsilon_t - 1 + \mu + \epsilon_t - 1 = \mu + \epsilon_t - 1 + \mu + \epsilon_t - 2 + \mu + \epsilon_t - 2 = \dots$$

$$\mu - \epsilon_t$$

$$\sum_{i=1}^n (x_i - \mu)^2$$

$$i=1$$

2. Sentimental Analysis Algorithm:

Essentially, sentiment analysis or sentiment classification fall into the broad category of text classification tasks where you are supplied with a phrase, or a list of phrases and your classifier is supposed to tell if the sentiment behind that is positive, negative or neutral. Sometimes, the third attribute is not taken to keep its binary classification problem. In recent tasks, sentiments like "somewhat positive" and "somewhat negative" are also being considered.

Step 1: Read the Data frame. import pandas as pd. ...

Step 2: Data Analysis. Now, we will take a look at the variable “Score” to see if majority of the customer ratings are positive or negative. ...

Step 3: Classifying Tweets. ...

Step 4: More Data Analysis. ...

Step 5: Building the Model. ...

Step 6: Testing

3. Naive Bayes algorithm

Naive Bayes classifier calculates the probability of an event in the following steps:

- Step 1: Calculate the prior probability for given class labels
- Step 2: Find Likelihood probability with each attribute for each class
- Step 3: Put this value in Bayes Formula and calculate posterior probability.
- Step 4: See which class has a higher probability, given the input belongs to the higher probability class.

Mathematical equation for Naïve Bayes Algorithm :

$$P(h|D) = \frac{P(D|h) P(h)}{P(D)}$$

- $P(h)$: The probability of hypothesis h being true (regardless of the data). This is known as prior probability of h .
- $P(D)$: The probability of the data (regardless of the hypothesis). This is known as prior probability.
- $P(h/D)$: The probability of hypothesis h is given the data D . This is known as posterior probability.
- $P(D/h)$: The probability of data d given that the hypothesis h was true. This is known as posterior probability.

Where:

$P(h/D)$ = posterior probability

$P(D/h)$ = likelihoods $P(h)$ = class prior probability

$P(D)$ = predictor prior probability

CHAPTER 07

07 Software Testing

7.1 Type of Testing

Purpose

Testing is an investigation conducted to provide stakeholders with information about the quality of the product or service under test. Software testing also provides an objective, independent view of the software to allow the business to appreciate and understand the risks of software implementation. Test techniques include, but are not limited to, the process of executing a program or application with the intent of finding software bugs. Software testing can also be stated as the process of validating and verifying that a software program or application or product:

1. Meets the business and technical requirements that guided its design and development;
2. Works as expected; and
3. Can be implemented with the same characteristics

Scope of Testing

Software testing, depending on the testing method employed, can be implemented at any time in the development process. However, most of the test effort occurs after the requirements have been defined and the coding process has been completed. As such, the methodology of the test is governed by the software development methodology adopted. Different software development models will focus the test

effort at different points in the development process. Newer development models, such as Agile, often employ test driven development and place an increased portion of the testing in the hands of the developer, before it reaches a formal team of testers. In a more traditional model, most of the test execution occurs after the requirements have been defined and the coding process has been completed.

Test Plan

To test this application, we are going with proper sequencing of testing like unit, integration, validation, GUI, Low level and High-level test cases, major scenarios likewise. We will go with the GUI testing first and then integration testing. After integration testing performs the high-level test cases and major scenarios which can affect the working on the application. We will perform the testing on the data transmitted using the various inputs and outputs and validate the results. It also intends to cover any deviations that the project might take from the initially agreed Test Strategy in terms of scope, testing methodology, tools, etc.. This test plan covers details of testing activities for this project and scope.

Software to be tested

1. Edraw Max:

It enables students, teachers and business professional store liable create and publish various kinds of diagram store present any ideas. With this application users can easily create professional- looking flow charts, organizational charts, network diagrams, business presentations, building plans, mind maps, science illustration, fashion designs, UML diagrams and much more.

Star UML:

Star UML is a fully fledged, open source, UML modeling tool that supports the ability to create software designs, from basic concepts, through to the coded solution. The user should be aware that this tool is

more complex than a simple UML diagram editing tool, in that, through the use of the model Drive Architecture (MDA) standard, the tool supports complex modeling which is realizable in code.

Test Cases

- GUI Testing

Graphical User Interface (GUI) testing is the one of the mechanisms in which user interface developed System Under some graphical rules. GUI testing includes checking various controls- menus, buttons, icons, dialog boxes and windows etc. Proposed system is tested for user inputs against different modules, validations are done. GUI is tested for appearance of different controls; visibility graphs is tested. GUI testing involves following actions:

1. Check all elements for size, position, width, length and acceptance of characters or numbers. For instance, you must be able to provide inputs to the input fields.
2. Overall functionality related with performance of user's graphical interface are checked.
3. Check Error Messages are displayed correctly
4. Check the font, layout details, style and display warning messages if it is false.
5. check the positioning of GUI elements.

- Unit Testing

It is the testing of individual software units of the application .it is done after the completion of an individual unit before integration. Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. This is a structural testing, that relies on knowledge of its construction and is invasive.

- Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

- Integration Testing

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfactory, as shown by successfully unit testing, the combination of components is correct and consistent.

Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

- Testing Strategy

Software testing methods are traditionally divided into white- and black-box testing. These

two approaches are used to describe the point of view that a test engineer takes when

designing test cases.

1. White-box testing

In white-box testing an internal perspective of the system, as well as programming skills, are used to design test cases.

2. Black-box testing

Black-box testing treats the software as a "black box", examining functionality without any

knowledge of internal implementation. The testers are only aware of what the software is supposed to do, not how it does it.

3. Grey-box testing

Grey-box testing involves having knowledge of internal data structures and algorithms for purposes of designing tests, while executing those tests at the user, or black-box level. The tester is not required to have full access to the software.

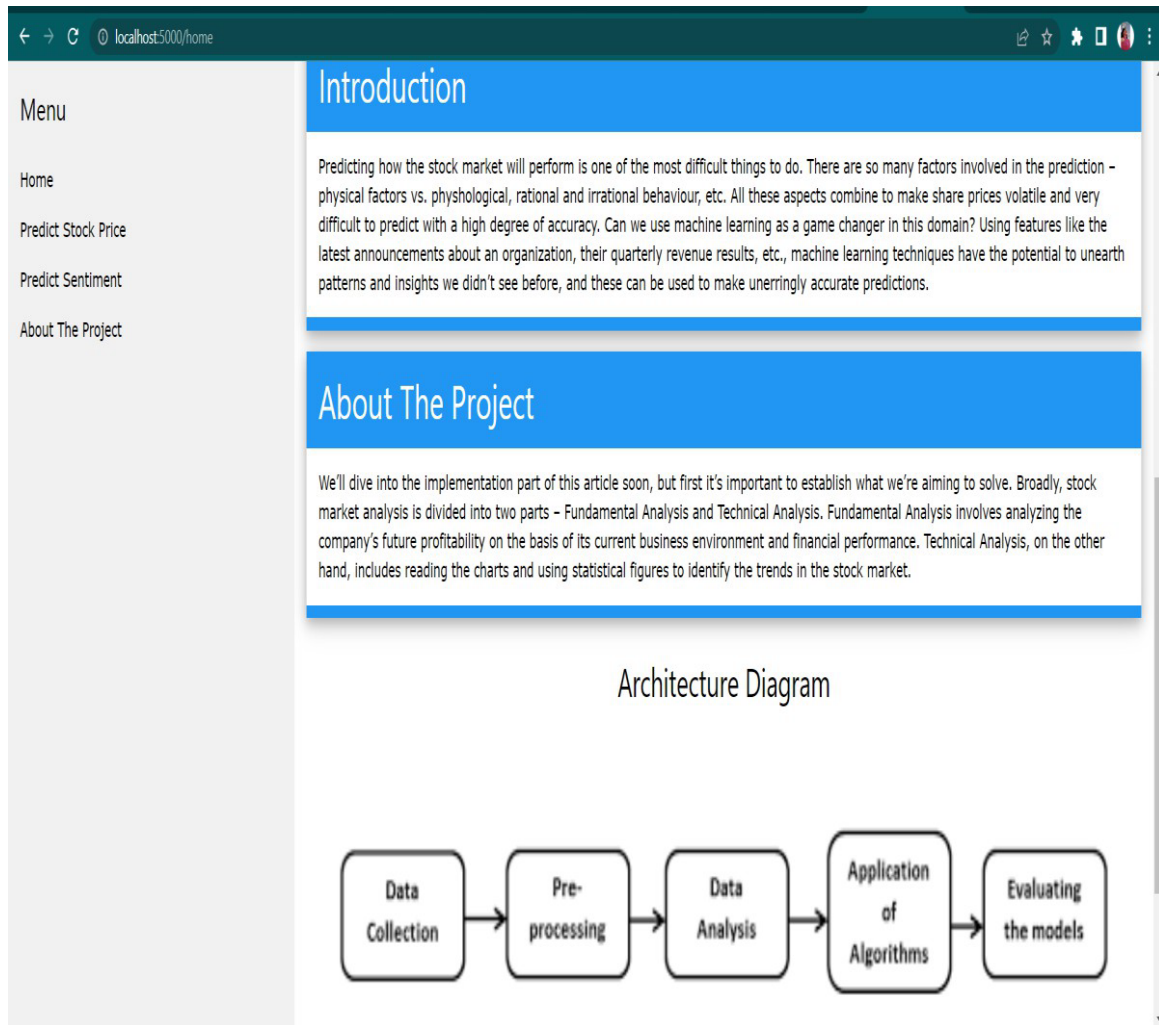
CHAPTER 08

08 Results

8.1 Outcomes

The screenshot displays a web application interface for 'Stock Price Prediction Using Machine Learning'. The browser's address bar shows 'localhost:5000'. On the left, a grey sidebar contains a 'Menu' with links to 'Home', 'Predict Stock Price', 'Predict Sentiment', and 'About The Project'. The main content area has a teal header with the application title. Below this is a blue 'Login' box containing a 'Username' field with the text 'admin', a 'Password' field with masked characters '*****', and a 'Login' button. The footer is teal and lists the development team: 'Developed By' (Bushra Naz Sayyed, Sanjana Pisal) and 'Guided By' (Prof. J.y.Kapadnis, Shaunak Deokuthe, Yash Laddha).

Login Page



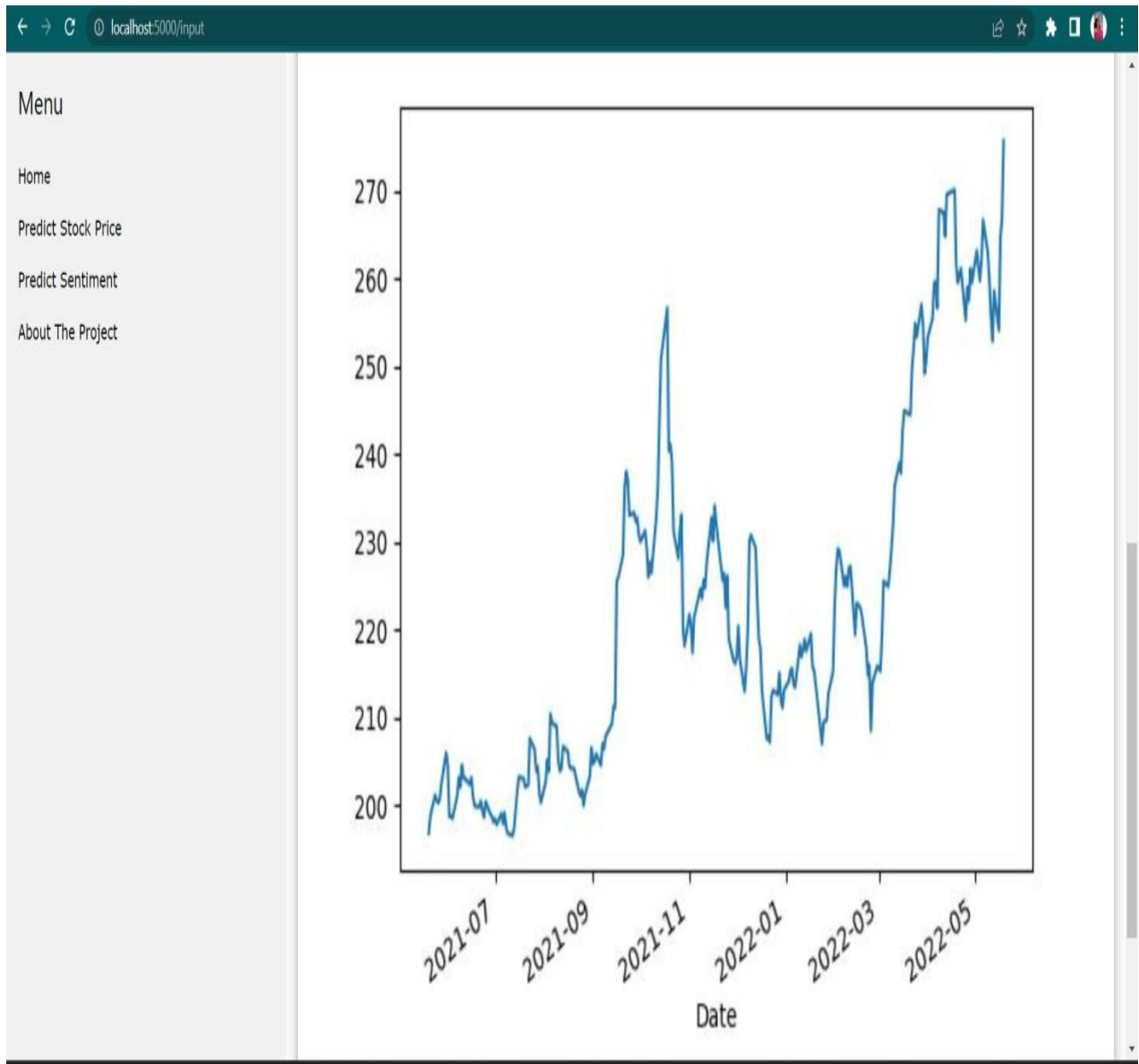
Introduction

The screenshot shows a web browser window at localhost:5000/input. The application has a sidebar menu with the following items: Menu, Home, Predict Stock Price, Predict Sentiment, and About The Project. The main content area is titled "Stock Price Prediction Using Machine Learning" and contains a "Stock Price Prediction" section. This section includes the following fields and controls:

- Enter Name of Stock:** A text input field containing "ITC.NS".
- Enter Start Date(YYYY-MM-DD):** A date input field containing "2021-05-20".
- Enter End Date(YYYY-MM-DD):** A date input field containing "2022-05-20".
- Predict:** A button to trigger the prediction.
- Current Price of Stock:** A text input field displaying "275.6499938964844".
- Predicted Price of Stock:** A text input field displaying "[276.4931893293555]".

The Windows taskbar at the bottom shows the date and time as 12:49 PM on 5/20/2022, along with various system icons.

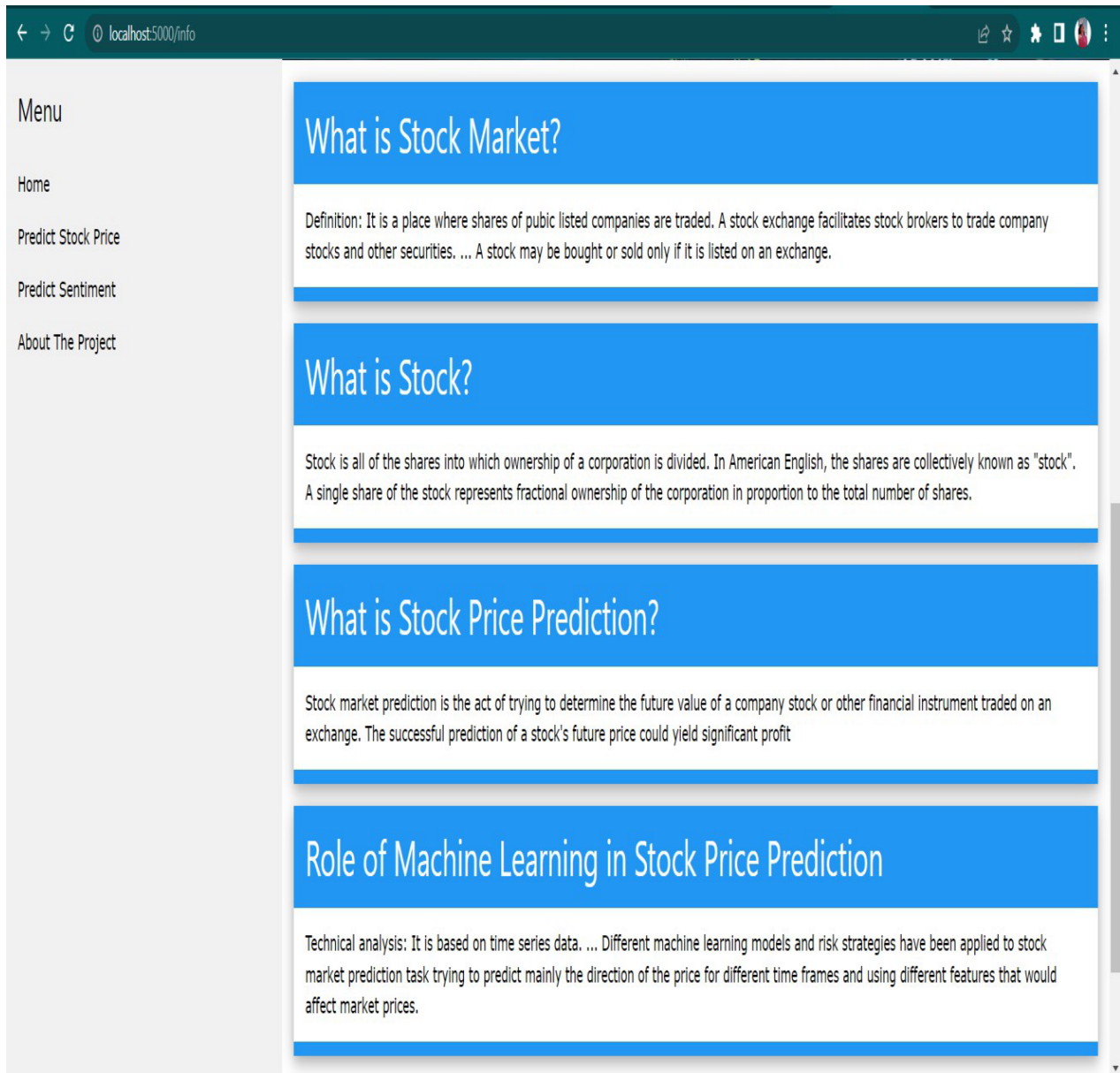
Stock Price Prediction



Analysis Report

The screenshot shows a web browser window with the address bar displaying 'localhost:5000/testing'. The browser's tab bar contains several open tabs, including 'Google', 'Sanju doc...', 'ssssss (1).d...', 'Meet...', '(78) Whats...', 'Final_Proje...', 'New Tab', 'partial rep...', 'Predicting...', 'Stock', and a plus sign for more tabs. The web application has a green header with the title 'Stock Price Prediction Using Machine Learning'. On the left, there is a grey sidebar menu with the following items: 'Menu', 'Home', 'Predict Stock Price', 'Predict Sentiment', and 'About The Project'. The main content area has a blue header for 'Sentiment Analysis'. Below this, there are several input fields with labels in green text: 'Stock Name' (containing 'axis.ns'), 'Total Tweets' (containing '0'), 'Positive Tweets' (containing '0'), 'Negative Tweets' (containing '0'), '% Price Increase' (containing '0'), and 'Prediction' (containing 'Stock Will Decrease').

Sentimental Analysis



About The Project

CHAPTER 09

09 Conclusions

9.1 Conclusions

The purpose of this study is to extend the accuracy to forecast the prediction of stock victimization the mixture of sentimental analysis, stock analysis and deep learning. the final word goal was to extend the yield from the investment. The accuracy was found to lean towards preciseness with increased information. last, we will say that if we tend to train our system with a lot of computer file set, modify the deep learning module to perform a lot of trial and errors tests, it's the potential to come up with a lot of correct and considerably a lot of consistent smallest error predictions.

9.2 Future Scope

- Live chart of a particular stock.
- Displaying the tweets of a selected stock.
- Show an inexperienced signal if stock is optimistic and red signal if the stock is pessimistic on the analysis chart.
- Show the stock price of a particular stock 24hr before prediction.

9.3 Applications

Stock worth Prediction victimization machine learning helps you discover the longer-term worth of company stock And different money assets listed on an exchange. the complete plan of predicting stock costs is to achieve important profits.

- We can use our application in society
- Industrial use
- Investors will use our application

Chapter 10

Appendix

Appendix A: Problem statement feasibility assessment using, satisfiability analysis and NP Hard, NP-Complete or P type using modern algebra and relevant mathematical models.

A correct prediction of stocks can lead to huge profits for the seller and the broker. Frequently, it is brought out that prediction is chaotic rather than random, which means it can be predicted by carefully analyzing the history of respective stock market. Machine learning is an efficient way to represent such processes. It predicts a market value close to the tangible value, thereby increasing the accuracy. Introduction of machine learning to the area of stock prediction has appealed to many researches because of its efficient and accurate measurements. The vital part of machine learning is the dataset used. The dataset should be as concrete as possible because a little change in the data can perpetuate massive changes in the outcome.

Feasibility Study:

All the projects are feasible if unlimited resources and unlimited time is given. Practically, it is not possible to provide unlimited resources and unlimited time to any project. The development of the computer based System is more likely to be affected by lack of resources and time deadlines Feasibility study is second stage in the “System Development Life Cycle”. According to result of preliminary investing examine the system feasibility & like hood that the System will be beneficial to organization. For feasibility some points are considered like: Any other creation is available for

1. Technical Feasibility:

In recent days security system has many disadvantages which is now generally preferred. Gesture authentication system is now more efficient and effective than previous systems. There is a need of related object and software, so all this technical part should be considered in this system.

2. Economic Feasibility:

Any system newly developed needs some initial investment. These are concerned with the cost of incurred for the development & implementation of software product. The maintenance of software product is derived from it the cost involved is that of coding, implementation of software product hence the system is economically feasible.

3. Operational Feasibility:

The system will be easy to use. The system will perform all the validation will be very user friendly & easy to understand for the user. The main user of the software will be a person who needs to have their information shared. The proposed system being Android Operating System based software is easy to operate and user friendly. As the software incorporate an intuitive interface, user needs to have skills related to sharing uploading and viewing on Mobile and PC.

NP Hard, NP-Complete or P type using modern algebra and relevant mathematical models.

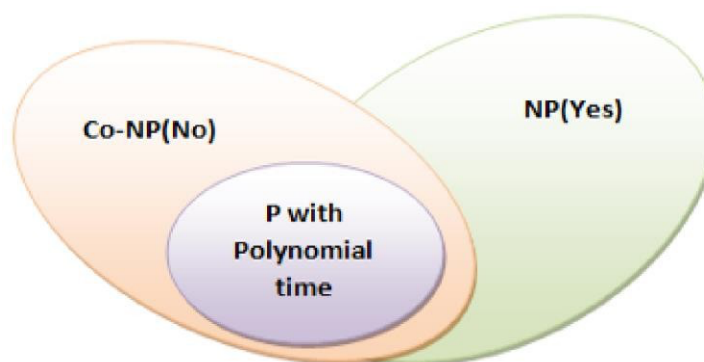
Np-hard Np- Complete:

What is P? _ P is set of all decision problems which can be solved in polynomial time by a deterministic.

- Since it can be solved in polynomial time, it can be verified in polynomial time.
- Therefore, P is a subset of NP.

P:

- In Existing system, Stock Price Prediction is very helpful for investors as well as for the applicant also. The aim of this system is to provide quick, immediate and easy way to choose the deserving applicants. It can provide special advantages to the Company. Various machine learning algorithms are used for stock data set and the objective is to predict the stock market.



What is NP?

"NP" means "we can solve it in polynomial time if we can break the normal rules of step-by-step computing".

What is NP Hard?

A problem is NP-hard if an algorithm for solving it can be translated into one for solving any

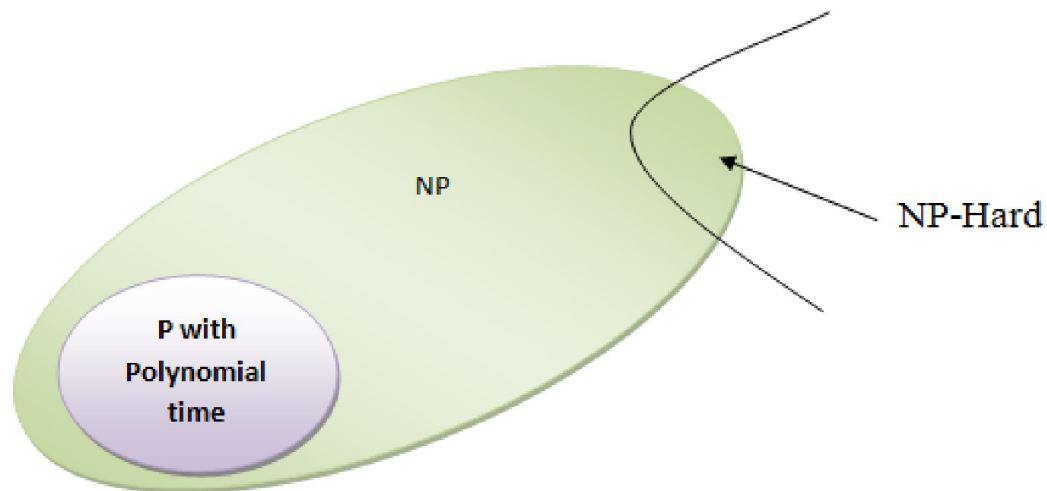
NP problem

(Non deterministic polynomial time) problem. NP-hard therefore means "at least as hard as any NP-problem," although it might, in fact, be harder.

NP-Hard:

So here in this case the 'P' problem is NP hard.

i.e., $P=NP\text{-Hard}$

**What is NP-Complete?**

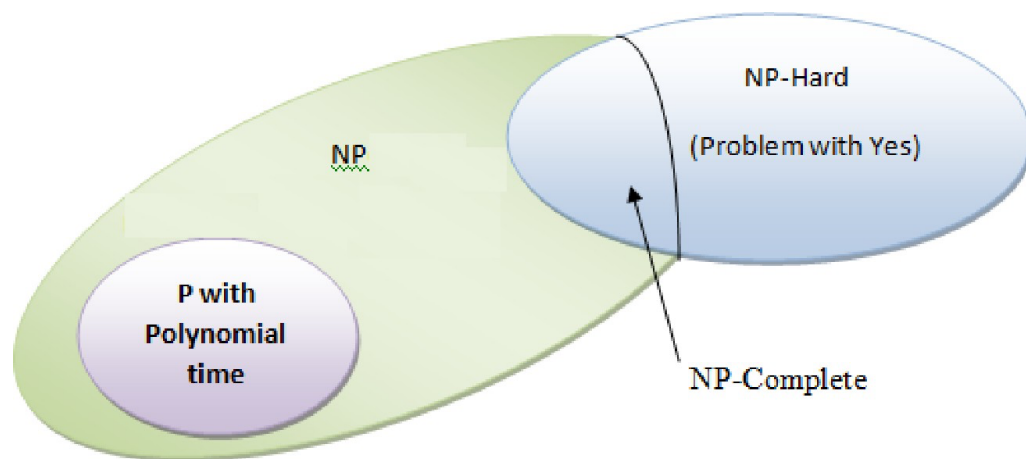
Since this amazing "N" computer can also do anything, a normal computer can, we know that "P" problems are also in "NP". So, the easy problems are in "P" (and "NP"), but the really hard ones are *only* in "NP", and they are called "NP-complete".

It is like saying there are things that People can do ("P"), there are things that Super People can do ("SP"), and there are things *only* Super People can do ("SP-complete").

NP-Complete:

We have used PYTHON language for coding.

Hence the 'P' is NP-Complete in this case.



Appendix B: Paper Publications And Competitions.

Name of Conference/journal	International Journal of Research in Engineering and Science (IJRES)
Detail of paper	ISSN (Online): 2320-9364, ISSN (Print): 2320-9356 www.ijres.org Volume 10 Issue 4 2022 PP. 21-26
Name of project competition	"PROJECT MANIA 2K22"- Sinhgad Academy of Engineering Kondhwa (Bk), Pune-48

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Predicting Stock Market trends using Machine Learning and Deep Learning

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Abstract: Equity market forecasting has long been an area of interest to investors and researchers due to volatility, complexity and the ever-changing nature of, making it difficult to reliably predict. Machine learning (ML) and data mining are based on the theory that historical data contains important storage for predicting the future direction. This technology is designed to help investors discover hidden patterns of historical data that are likely to be predictable when making investment decisions. Forecasting stock market is considered a difficult task in financial time series forecasting using ARIMA model. Data analysis is a way to predict whether future stock prices will rise or fall. We also looked at various global events and forecasts for the stock market. The stock market can be seen as a special data mining topic. Text mining approaches are also used to measure the impact of real-time news on stocks. Predict the ups and downs of the stock market using a variety of techniques and strategies.

Keywords: given to the voice recognition features are as follows:-
Forecasting, Data analysis, Prediction, Text mining

Date of Submission: 06-04-2022

Date of acceptance: 22-04-2022

I. INTRODUCTION

Rapid advances in digital data capture are rapidly increasing the amount of data stored in databases, data warehouses, and other types of data storage. Data can hide valuable information, but the vast amount of data makes it difficult for humans to extract without powerful tools. Until the beginning of the last decade, news was not available easily and quickly. In this information age, content providers and content locators, such as online news services, have emerged on the World Wide Web, making news easily accessible. The constant availability of more news articles in digital format, the latest developments in natural language processing (NLP), and the availability of faster computers, how more information can be extracted from news articles. It leads to the problem of. Financial analysts investing in the stock market are usually unaware of stock market behavior. They face stock trading problems because they don't know which stocks to buy and which stocks to sell in order to make more profits. All these users know that the development of the stock market depends heavily on the relevant news and needs to process vast amounts of information on a daily basis. You need to analyze all the news that appears in newspapers, magazines, and other text sources. However, analyzing such an amount of financial news and articles to extract useful knowledge is beyond human ability. Text mining technology helps to automatically extract useful knowledge from text resources. We will use text mining technology to develop a system that can model and predict the reaction of the stock market to news articles. This allows investors to anticipate future behavior of the stock and take immediate action when relevant news is released. Use real-time news articles and daytime stock prices from several companies on the Bombay Stock Exchange as input.

The overall purpose of the survey can be summarized in the following survey questions:

- How can you use textual financial news to predict stock price reactions?
- How can data and text mining techniques help generate this predictive model?

To investigate the impact of news on the movement of stock trends, we need to build a forecast model.

Predicting the development of the stock market is one of the most difficult tasks. There are numerous factors involved in forecasting. Physical factors and psychological, rational and irrational behavior. All of these factors can cause stock prices to fluctuate, making it extremely difficult to predict accurately.

*Predicting Stock Market trends using Machine Learning and Deep Learning***II. MOTIVATION**

Predicting how the inventory marketplace works is one of the maximum hard tasks. There are several elements concerned in prediction- bodily . All those elements integrate to make percentage fees risky and really difficult to expect with a excessive diploma of accuracy. Predicting inventory marketplace moves is a famous and thrilling topic. Today, social media is a whole expression of public opinion and critiques approximately modern events. Twitter mainly attracted the eye of researchers reading public opinion. Opinion-based market forecasting expressed on Twitter is a fascinating area of research. A previous study concluded that the general public opinion collected by Twitter correlates well with the Dow Jones Industrial Average (DJIA). The purpose of this paper is to observe how the volatility, rise and fall of a company's stock price correlates with the 4,444 public opinion expressed in a tweet about that company. Understanding the author's opinion from the text is the goal of sentiment analysis.

III. OBJECTIVE

To predict the future value of the financial stocks of a company. To implement prediction technologies with use of machine learning which makes prediction based on the values of current stock market. To use twitter data for prediction via sentimental analysis to increase the efficiency of system. To achieve maximum accuracy.

IV. LIMITATION

Continues requirement of Internet connection. Accurate Input
Data Required. For sentimental analysis proper segregation of the acquired data is required.

V. LITERATURE SURVEY

This section describes related work. We have identified related work in two different areas: technology and finance. Kim and Han constructed a discretized model of the function for predicting stock indexes as a combination of artificial neural networks (ANN) and genetic algorithms (GA). The data used in their study include technical indicators and the direction of daily changes in the Korea Stock Index (KOSPI). They used data containing a sample of 2928 trading days from January 1989 to December 1998 to report selected characteristics and formulas. They also applied feature discretization optimization as a technique similar to dimensionality reduction. The strength of their work is the implementation of GA to optimize ANN. First, the number of hidden layer input features and processing elements is 12, which cannot be adjusted. Another limitation lies in the learning process of ANN, where the author focuses only on two elements of optimization. Although they still believed that GA had great potential for optimizing the discretization of functions. The initialized feature pool is related to the selected feature. In this work, the author uses a genetic algorithm with a model based on an artificial neural network, which he calls a hybrid GA-ANN model. Piramuthu has conducted a thorough evaluation of various feature selection methods for data mining applications. He used datasets that are loan approval data, loan default data, web traffic data, tom and kian data, and compared how various feature selection methods optimize the performance of decision trees. Among the methods he compared with the features he chose were probabilistic distance measures classes: Minkovsky distance measures, city block distance measures, Euclidean distance measures, Chebisev distance measures, non-linear (Parzen and Hypersphere kernel) distance measures. Hassan and Nath applied Hidden Markov Model (HMM) to stock market forecasts. Stock prices of four different airlines. Reduces the state of the model to four states: open price, close price, highest price, and lowest price. The strength of this white paper is that this approach does not require expertise to build a predictive model. This task is limited within the aviation industry and is interpreted on very small datasets, but it may not be a predictive model with general validity. One of the working approaches to stock market forecasting can be used for comparison work. The author selected up to 2 years as the date range for the training and testing dataset. This provided a reference to the date range of the review. Lei used Wavelet Neural Network (WNN) to predict stock price trends. For optimization, the author also uses an approximate set (RS) to reduce the attribute. Approximate sets are used to reduce the dimension of stock price trends. It has also been used to determine the structure of wavelet neural networks. The dataset for this work includes five well-known stock indices (1) SSE Composite Index (China), (2) CSI 300 Index (China), (3) All Common Indexes (Australia), (4) It is composed of Nikkei 225 Index (Japan) and (5) Dow Jones Index (USA). The model's valuation is based on various stock market indices, and the results were convincing because of its general validity. Using rough sets to optimize feature dimensions before processing reduces computational complexity. However, in the discussion part, the author emphasized only parameter adjustments and did not identify weaknesses in the model itself. In the meantime, I also noticed that the index was being valued. If you apply the same model to a particular stock, you may not get the same performance. Sirignano and Cont were used in a deep learning solution trained on the universal functions of financial markets. The dataset used consisted of trading records for all transactions and cancellation of orders for approximately 1000 NASDAQ shares via stock exchange purchase orders. NN consists of three layers with LSTM units and finally a lookahead layer with rectified linear unit (ReLU), using a stochastic gradient descent (SGD) algorithm for optimization.

Predicting Stock Market trends using Machine Learning and Deep Learning

Their universal model can generalize and cover constraints other than those of the training data. They mentioned the benefits of the universal model, but the cost of training was still high. In the meantime, it's unclear if the mysterious programming of deep learning algorithms contaminates useless functions when data is fed into the model. The authors found that it was better to perform the feature selection part before training the model, and found that this was an effective way to reduce computational complexity.

VI. ARCHITECHTURE DAIGRAM

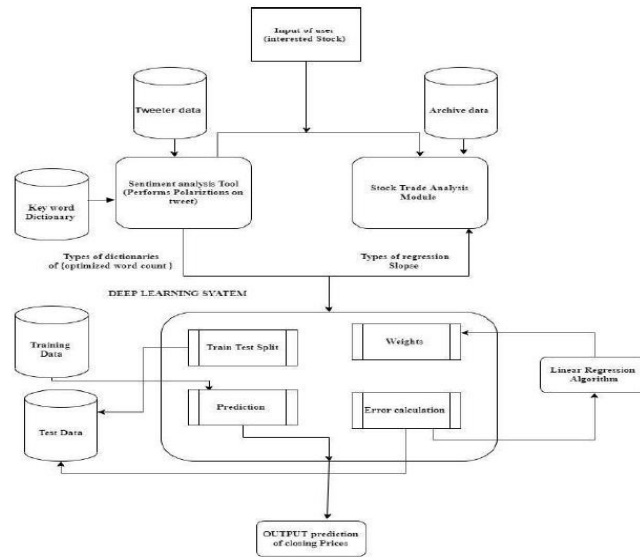


Figure1: System Architecture

Accurately predicting stocks can bring enormous profits to both sellers and brokers. It is often pointed out that predictions are more chaotic than random. This means that you can make predictions by carefully analyzing the history of the stock market in question. Machine learning is an efficient way to map such processes. Predict market value close to material value, thereby increasing accuracy. The advent of machine learning in the field of inventory forecasting has fascinated many researchers for its efficient and accurate measurements as small changes in the data can persist large changes in the results. In this system, supervised machine learning is applied to datasets supplied by Tweeter Finance. This dataset contains five variables: Open, Close, Low, High, and Volume. Open, close, low, and high prices are different bid prices for stocks at different points in time that have almost direct names. Volume is the number of shares passed from one owner to another during the period. The model is then tested against the test data.

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VII. PROJECT IMPLEMENTATION: Module 1 (Login Module)

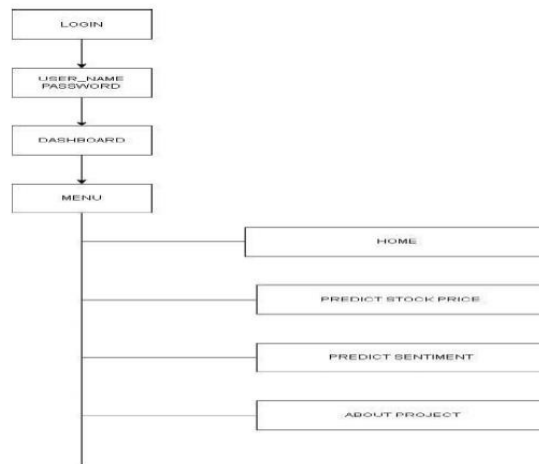


Figure 2: Login Module

The Login Module Consists of Dashboard Where the user can login into the account and will be directed to the homepage.

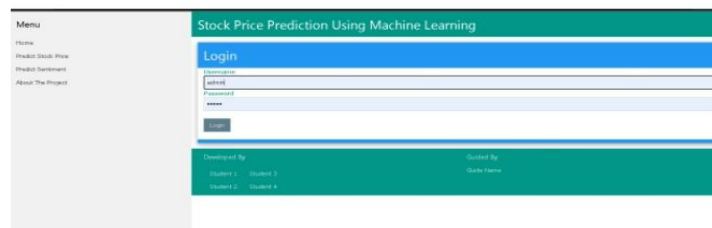
The Menu option consists of Home, Predict stock price, Predict Sentiment and About the project.

Home: The home Page Consists of the introduction part of the topic opted for project i.e. Information regarding the stock market.

Predict Stock Price: This page consists of the prediction part. The price of the stocks are predicted here with the entry of stock name, start date and end date.

Predicted sentiment: On this page the sentiment analysis of the stock is done and the increase or decrease in the price is predicted.

About the Project: Here the details about the project is displayed along with Architecture Diagram.



VIII. Module 2 (ARIMA)

1. Auto Regressive Integrated Moving Average (ARIMA)

- A time series forecasting model is a model capable of predicting future values based on previously observed values.

- Time series forecasting is widely used for non-stationary data.

- Non-stationary data is called data whose statistical properties, such as mean and standard deviation, are not fixed over time, but instead, these measurements change over time.

Breaking down these terms we get:

1) AR: means that the model uses a dependency between the observations and a predefined number of delay observations. (Also known as "time lag" or "delay").

2) I: means that the model uses the derivative of the raw observations of

Predicting Stock Market trends using Machine Learning and Deep Learning

3) MA: means that the model utilizes the relationship between residual error, residual error and observed values. Steps for implementation are as follows:

1. Visualize the time series data.
2. Make the time series data stationary.
3. Plot the correlation and auto-correlation chart.
4. Construct the ARIMA model based on the data.
5. Use the model to make predictions.

Mathematical Equation For ARIMA is as follows :

$$z_t = a_t + 1 - a_t$$

$$z_t = \phi_1 z_{t-1} + \theta_1 \varepsilon_{t-1} + \varepsilon_t$$

Recover ak

$$ak = z_{k-1} + ak - 1 = z_{k-1} + z_{k-2} + ak - 2$$

=...

$$k-1$$

$$\sum_{i=1}^{k-1} z_{k-i+1}$$

$$i=1$$

IX. Module 2 (Sentimental Analysis)

Sentiment analysis or emotion classification is broadly divided into a wide range of text classification tasks. It displays a sentence or a list of sentences, letting the classifier know if the emotion behind it is positive, negative, or neutral. To maintain the problem of binary classification, the third attribute may not be retrieved. For current tasks, moodssuch as "quite positive" and "quite negative" are also taken into account.

Step 1: Read the Dataframe. import pandas as pd. ...

Step 2: Data Analysis. Next, let's look at the variable "score" to see if most customer reviews are positive or negative.

Step 3: Classifying Tweets.

Step 4: More Data Analysis.

Step 5: Building the Model

Step 6: Testing

Naive Bayes algorithm

The Naive Bayes classifier calculates the probability of an event by doing the following:

- Step 1: Calculate prior probabilities for a particular class label
- Step 2: Find the probability probability for each attribute of each class
- Step 3: Substitute these values into Bayesian equations to calculate posterior probabilities.
- Step 4: If the input belongs to a class with a higher probability, find out which class has the higher probability.

Mathematical equation for Naïve Bayes Algorithm:

$$P(h|D) = \frac{P(D|h)P(h)}{P(D)}$$

$$P(D)$$

- $P(h)$: The probability of hypothesis h being true (regardless of the data). This is known as prior probability of h.
- $P(D)$: The probability of the data (regardless of the hypothesis). This is known as prior probability.
- $P(h|D)$: The probability of hypothesis h is given the data D. This is known as posterior probability.
- $P(D|h)$: The probability of data d given that the hypothesis h was true. This is known as posterior probability.

X. NLP (Natural Language Processing)

Natural Language Processing (NLP) refers to the field of computer science, especially artificial intelligence or AI. This allows computer systems to understand text and spoken language as humans do. NLP combines complete rule-based modeling of human language in computational linguistics with statistics, device-related, and deep learning modes. The combination of these technologies allows computer systems to systematize human audio in the form of textual or audio data and "understand" the whole, the purpose and

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emotion of the speaker or creator. NLP drives a laptop package that translates text content from one language to another, responds to voice commands, and quickly summarizes large amounts of text content in real time. It is very likely that you have interacted with NLP in the form of voice-managed GPS structures, virtual assistants, voice reputation dictation software, customer support chatbots, and other customer conveniences. However, NLP also plays a developmental role in the organization's response to streamline business operations, increase employee productivity, and simplify critical business processes. Laptop algorithms, combined with modal learning and deep learning devices, mechanically extract, classify, and label

factors from textual content and audio data, statistically all possible means of these factors. To capture. Enter the statistic NLP that provides the gender. The NLP structure is very raw, unstructured, untagged, and in-situ. I have. You can "learn" with you. You can extract more detailed information from the text content. And voice recording.

XI. FUTURESCOPE

- Livechart of a particular stock.
- Displaying the tweets of a particular stock.
- Show green signal if stock is bullish and red signal if the stock is bearish on the analysis chart.
- Show the stock price of a particular stock 24hrs before prediction.

XII. CONCLUSION

The purpose of this study is to use a combination of sentiment analysis, stock trend analysis, and deep learning to improve the accuracy of stock forecasting. The ultimate goal was to increase the rate of return on investment. The accuracy was found to lean towards precision with increased data. In conclusion, we can say that if we train our system with more input data set, modify the deep learning module to perform more trials and error tests, it has the potential to generate more accurate and significantly more consistent minimal error predictions.

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Appendix C: Plagiarism Report of project report.

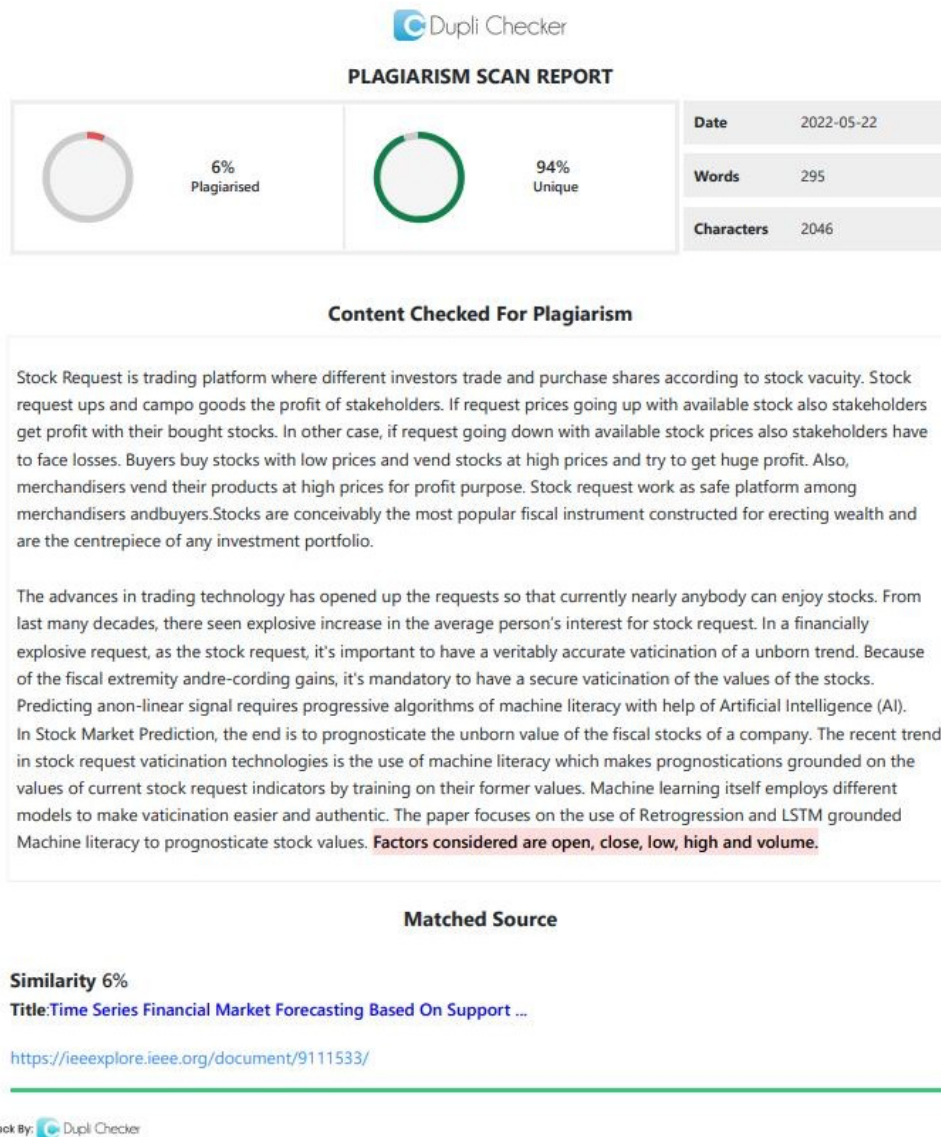
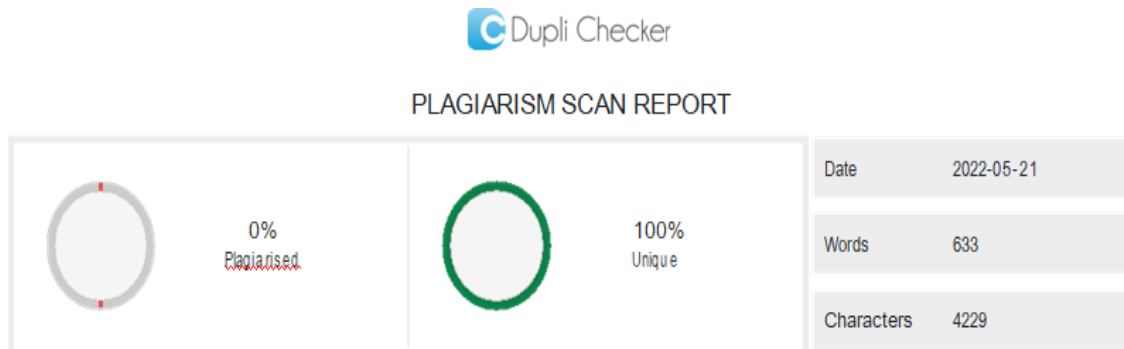


Fig 1. ABSTRACT



Content Checked For Plagiarism

The fast progress in digital information acquisition has crystal rectifier to the invasive quantity of knowledge hold on in databases, information warehouses, or different kinds of knowledge repositories. though valuable info could also be concealment behind the info, the overwhelming information volume makes it tough for citizenry to extract them while not powerful tools. straightforward and fast availableness to news info wasn't attainable till the start of the last decade. during this age of data, news is currently simply accessible, as content suppliers and content locators like on-line news services have up on the planet Wide net. Continuous availableness of additional news articles in digital type, the newest developments in linguistic communication process (NLP) and therefore the availableness of quicker computers cause the question the way to extract additional info out of reports articles.

Financial analysts United Nations agency invest available markets typically don't seem to be attentive to the exchange behavior. they're facing the matter of stock commercialism as they are doing not recognize that stocks to shop for and that to sell so as to achieve additional profits. of these users recognize that the progress of the exchange depends a great deal on relevant news and that they ought to deal daily with immense quantity of data. they need to investigate all the news that seems on newspapers, magazines and alternative matter resources. however analysis of such quantity of economic news and articles so as to extract helpful information exceeds human capabilities. Text mining techniques will facilitate them mechanically extracting the helpful information out of matter resources.

1.2 Motivation

In the existing system there's AN solely Current worth of stock value isdone. That mean we are able to take solely the front and back read of the item and store it.

Fig 2. INTRODUCTION

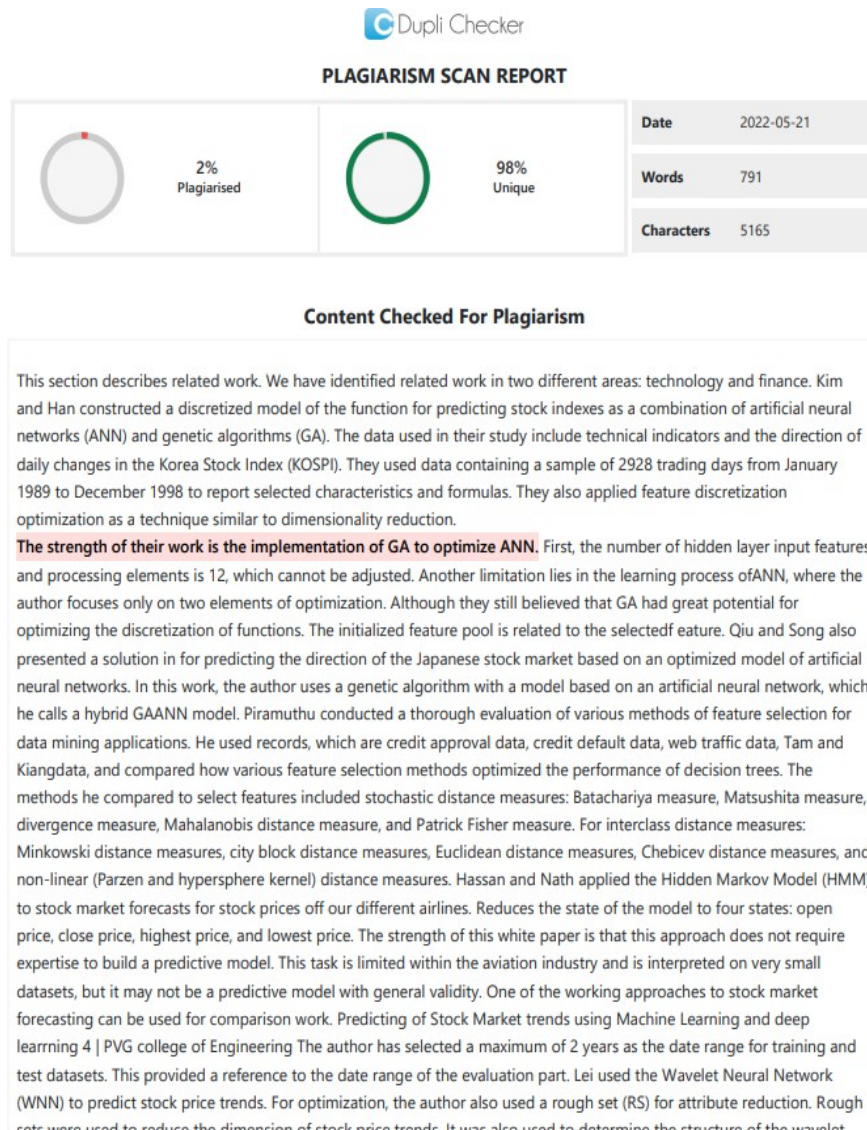


Fig 3. LITERATURE SURVEY

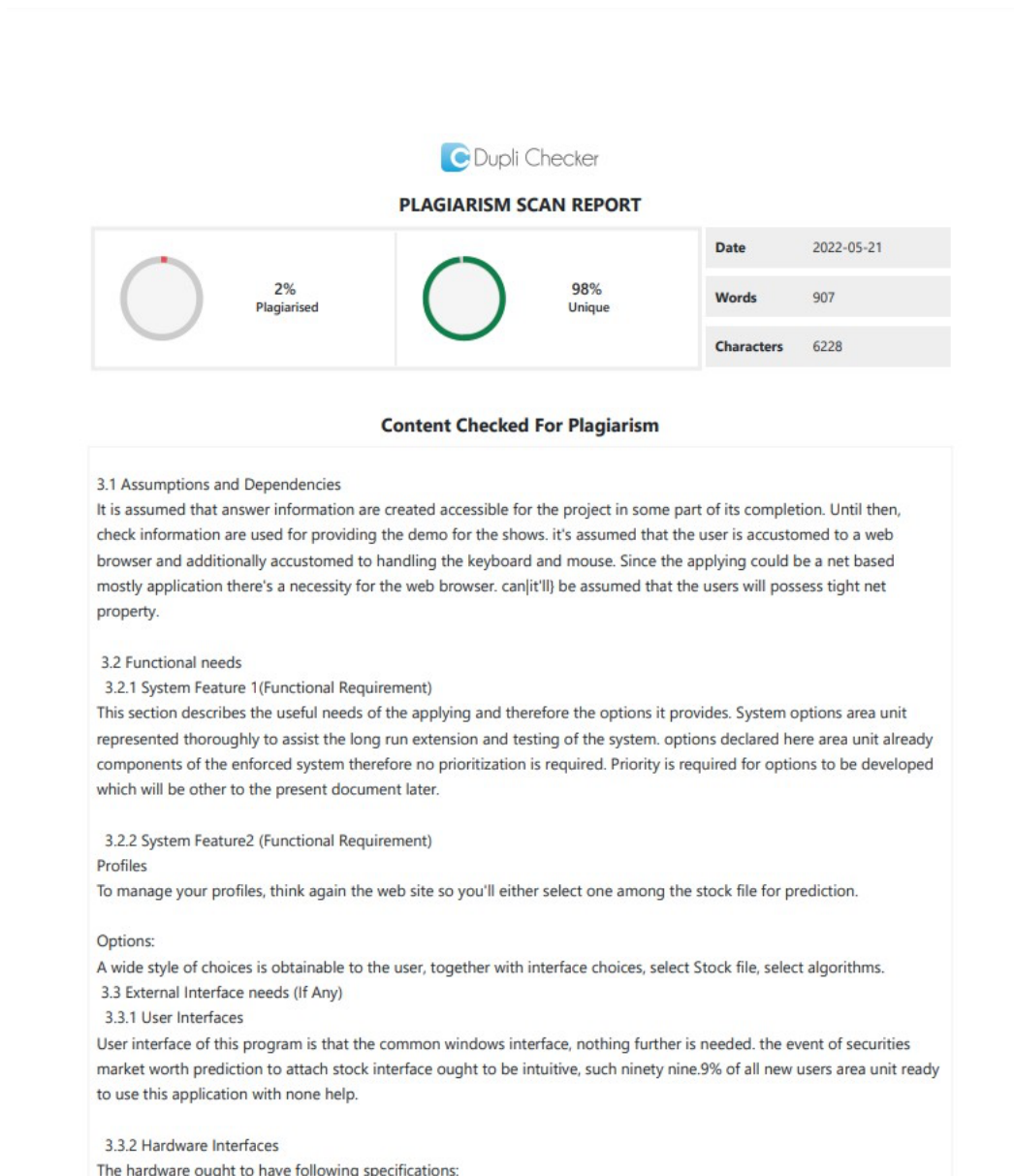


Fig 4. SYSTEM REQUIREMENTS AND SPECIFICATIONS

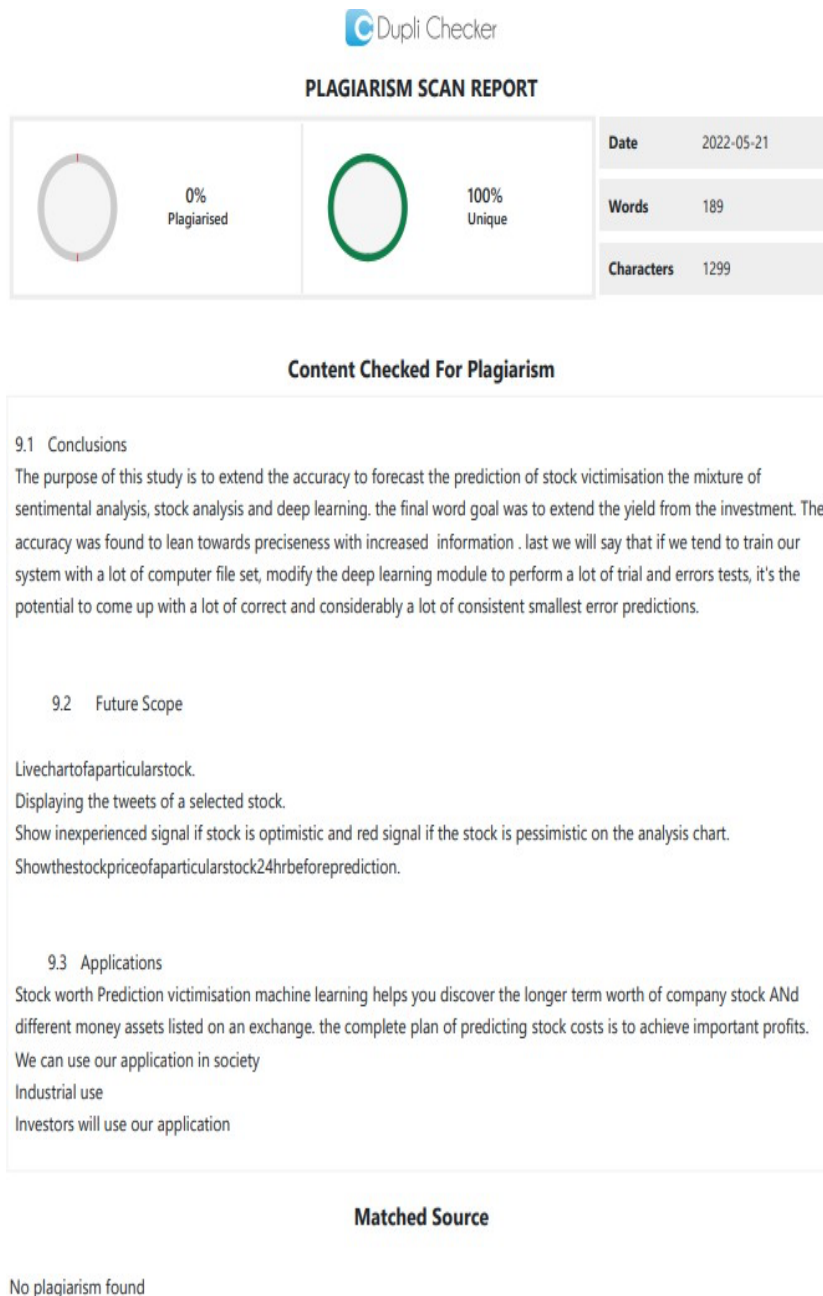


Fig 5. CONCLUSION

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